SET 47 • 7.G.B.6

BACKWARD 1

Find the volume of a rectangular prism with fractional edges • Grade 6 • Geometry • 6.G.A.2

A right rectangular prism is $2\frac{1}{2}$ feet by 2 feet by 4 feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q1/5

A cube has $\frac{3}{4}$-inch edges.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q2/5

A right rectangular prism is $\frac{1}{2}$ m by 3 m by 6 m.

What is its volume?

A. 9.5 cubic meters

B. 4.5 cubic meters

C. 9 cubic meters

D. 18 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q3/5

True or false?

A right rectangular prism $1\frac{1}{2}$ cm by 2 cm by 2 cm has a volume of 6 cubic centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q4/5

True or false?

A right rectangular prism $\frac{1}{2}$ foot by $\frac{1}{2}$ foot by 4 feet has a volume of 2 cubic feet.

A. True

B. False

SET 47 • 7.G.B.6

BACKWARD 2

Find volume with V = l × w × h • Grade 5 • Measurement & Data • 5.MD.C.5.b

A rectangular prism is 7 units long, 4 units wide, and 3 units tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q1/5

A box is 10 cm long, 6 cm wide, and 5 cm tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q2/5

A rectangular prism is 9 feet by 2 feet by 3 feet.

What is its volume?

A. 27 cubic feet

B. 14 cubic feet

C. 108 cubic feet

D. 54 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q3/5

True or false?

A cube with 4-inch edges has a volume of 64 cubic inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q4/5

True or false?

A box 5 m by 5 m by 2 m has a volume of 12 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 7.G.B.6

FORWARD 1

Compare the volumes of cylinders and prisms • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 4 inches and a height of 5 inches.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q1/5

A cylinder's base has an area of 12.56 square centimeters. The cylinder is 10 cm tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q2/5

A cylinder has a radius of 1 m and a height of 3 m.

What is its volume in terms of π?

A. π cubic meters

B. 9π cubic meters

C. 6π cubic meters

D. 3π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q3/5

True or false?

The volume of a cylinder is its base area times its height, just like a prism.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 2 and a height of 2 has a volume of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 7.G.B.6

FORWARD 2

Use volume formulas to solve problems • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A pyramid has a square base with 6 m sides. It is 10 m tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q1/5

A cone-shaped cup has a radius of 3 cm and holds 30π cubic centimeters.

How tall is the cup?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q2/5

A sphere has a volume of 288π cubic centimeters.

What is its radius?

A. 6 cm

B. 12 cm

C. 4 cm

D. 8 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q3**

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q3/5

True or false?

A pyramid has $\frac{1}{3}$ the volume of a prism with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q4/5

True or false?

A sphere with a radius of 3 has a volume of 27π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q5/5

SET 48 • 7.G.B.6 • GEOMETRY

Find the surface area of right prisms

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Use a net to find surface area	**6.G.A.4**
BACKWARD 2	Find the area of a rectangle	**4.MD.A.3**
FORWARD 1	Use the Pythagorean theorem in three-dimensional figures — near…	**8.G.B.7**
FORWARD 2	Apply geometric methods to solve design problems	**HSG-MG.A.3**

SET 48 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

Find the surface area of a right rectangular prism that is 6 cm by 4 cm by 3 cm.

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q1/5

Find the surface area of a cube with 5-inch edges.

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q2/5

A right triangular prism has triangular bases with sides of 3 cm, 4 cm, and 5 cm and a right angle between the 3 cm and 4 cm sides. The prism is 10 cm long.

What is its surface area?

A. 126 square centimeters

B. 60 square centimeters

C. 132 square centimeters

D. 120 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q3/5

True or false?

A box 10 in by 2 in by 2 in needs exactly 88 square inches of paper to cover it with no overlap.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q4/5

True or false?

A right rectangular prism 2 m by 3 m by 4 m has a surface area of 24 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q5/5

SET 48 • 7.G.B.6

BACKWARD 1

Use a net to find surface area • Grade 6 • Geometry • 6.G.A.4

The net of a cube has 6 squares. Each square is 4 cm by 4 cm.

What is the surface area of the cube?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q1/5

The net of a rectangular prism has two 5-by-3 rectangles, two 5-by-2 rectangles, and two 3-by-2 rectangles.

What is the surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q2/5

The net of a square pyramid has a 6-by-6 square and 4 triangles. Each triangle has a base of 6 and a height of 5.

What is the surface area?

A. 186 square units

B. 66 square units

C. 156 square units

D. 96 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q3/5

True or false?

A net shows all the faces of a solid laid flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q4/5

True or false?

The net of a rectangular prism has 4 rectangles.

A. True

B. False

SET 48 • 7.G.B.6

BACKWARD 2

Find the area of a rectangle • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 8 cm by 5 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q1/5

A rectangle is 14 inches by 3 inches.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q2/5

A rectangle is 11 m by 6 m.

What is its area?

A. 34 square meters

B. 66 square meters

C. 17 square meters

D. 60 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q3/5

True or false?

A rectangle 9 feet by 2 feet has an area of 18 square feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q4/5

True or false?

A rectangle 5 cm by 5 cm has an area of 20 square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 7.G.B.6

FORWARD 1

Use the Pythagorean theorem in three-dimensional figures • Grade 8 • Geometry • 8.G.B.7

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A box is 4 units long, 3 units wide, and 12 units tall.

How long is a diagonal of its 4-by-3 base?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q1/5 • NEAREST RELATED

A box is 4 units long, 3 units wide, and 12 units tall.

How long is the diagonal from a bottom corner to the opposite top corner?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q2/5 • NEAREST RELATED

A right square pyramid (its top vertex is directly above the center of its base) has base edges of 6 m and is 4 m tall.

What is the slant height (from the top vertex to the midpoint of a base edge)?

A. 10 m

B. 7 m

C. 5 m

D. 2 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A rectangle that is 6 by 8 has a diagonal of 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

A cube with edges of 1 has a diagonal from one corner to the opposite corner of length 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 48 • 7.G.B.6

FORWARD 2

Apply geometric methods to solve design problems • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A company wants a box shaped like a cube with a volume of 64 cubic inches.

What is the edge length, and how much cardboard covers the box?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q1/5

One gallon of paint covers 350 square feet. The walls of a room total 1,225 square feet.

How many whole gallons should be bought?

Gallons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q2/5

Each box has a volume of 24 cubic centimeters.

Which box uses the least cardboard?

A. 1 cm × 4 cm × 6 cm

B. 1 cm × 1 cm × 24 cm

C. 2 cm × 3 cm × 4 cm

D. 1 cm × 2 cm × 12 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q3/5

True or false?

Among rectangular boxes with the same volume, a cube has the least surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q4/5

True or false?

Two boxes with the same volume always have the same surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q5/5

SET 67 • 7.G.B.6 • GEOMETRY

Find the volume of a solid made of right prisms

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Add the volumes of two rectangular prisms	**5.MD.C.5.c**
BACKWARD 2	Find the area of a figure made of rectangles	**3.MD.C.7.d**
FORWARD 1	Find the volume of a solid made of a prism and a cylinder	**8.G.C.9**
FORWARD 2	Use volume formulas to solve problems with composite solids	**HSG-GMD.A.3**

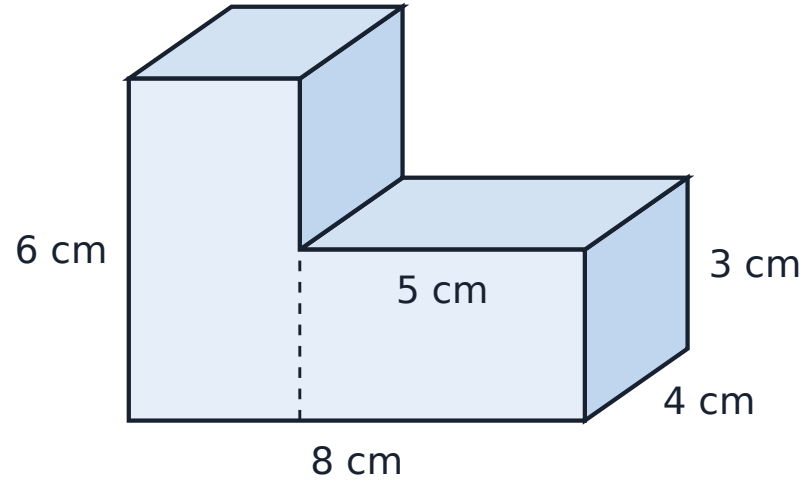
SET 67 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

The solid is made of two right rectangular prisms joined face to face with no gaps or overlaps.

Find its volume.



Volume: _____

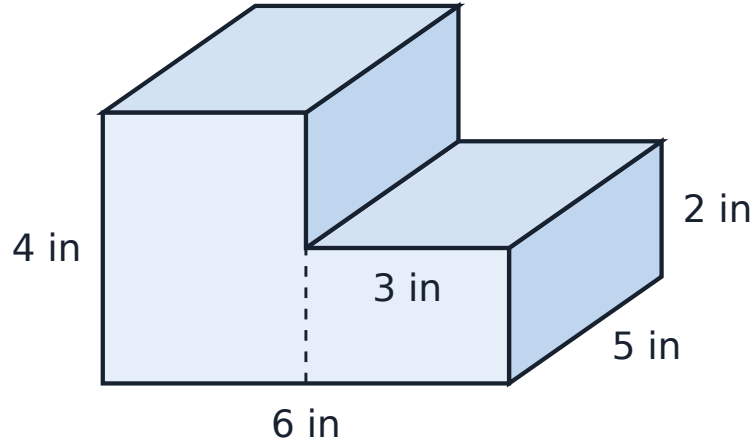
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 67 • MAIN • Q1/5

The step-shaped solid is made of two right rectangular prisms joined face to face with no gaps or overlaps.

Find its volume.



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 67 • MAIN • Q2/5

A cube with 4 cm edges sits on top of a right rectangular prism that is 8 cm by 6 cm by 2 cm. The bottom face of the cube lies flat on the top face of the prism.

What is the total volume?

A. 160 cubic centimeters

B. 64 cubic centimeters

C. 224 cubic centimeters

D. 96 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 67 • MAIN • Q3/5

True or false?

A solid is made of a 10 m by 4 m by 3 m prism and a 2 m by 4 m by 3 m prism joined end to end with no overlap. Its volume is 144 cubic meters.

A. True

B. False

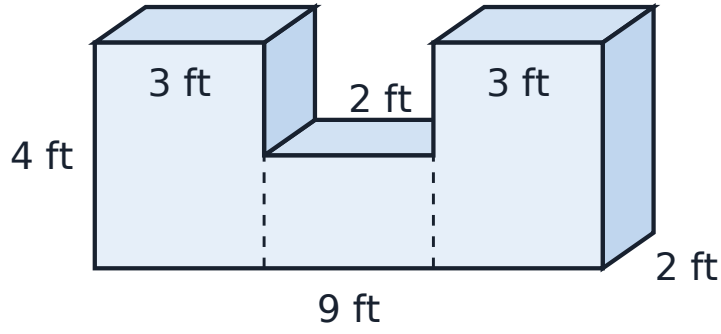
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 67 • MAIN • Q4/5

True or false?

The solid is made of three right rectangular prisms with no gaps or overlaps. Its volume is 72 cubic feet.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 67 • MAIN • Q5/5

SET 67 • 7.G.B.6

BACKWARD 1

Add the volumes of two rectangular prisms • Grade 5 • Measurement & Data • 5.MD.C.5.c

A 3 by 2 by 2 prism and a 4 by 2 by 1 prism are joined with no overlap.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 67 • BACKWARD 1 • Q1/5

A 5 by 3 by 2 prism and a 2 by 3 by 2 prism are joined with no overlap.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 67 • BACKWARD 1 • Q2/5

Two cubes with 3-unit edges are joined face to face.

What is the total volume?

A. 27 cubic units

B. 18 cubic units

C. 54 cubic units

D. 36 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 67 • BACKWARD 1 • Q3/5

True or false?

A prism with a volume of 10 cubic units and a prism with a volume of 6 cubic units are joined with no overlap. The total volume is 16 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 67 • BACKWARD 1 • Q4/5

True or false?

A 2 by 2 by 2 cube and a 2 by 2 by 4 prism are joined with no overlap. The total volume is 12 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 67 • BACKWARD 1 • Q5/5

SET 67 • 7.G.B.6

BACKWARD 2

Find the area of a figure made of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.d

A 4 by 2 rectangle and a 2 by 3 rectangle are joined with no overlap.

What is the total area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 67 • BACKWARD 2 • Q1/5

A 6 by 3 rectangle and a 3 by 3 rectangle are joined with no overlap.

What is the total area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 67 • BACKWARD 2 • Q2/5

A U-shape is made of three rectangles with no overlap: 2 by 4, 4 by 2, and 2 by 4.

What is the total area?

A. 20 square units

B. 16 square units

C. 32 square units

D. 24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 67 • BACKWARD 2 • Q3/5

True or false?

A 5 by 2 rectangle and a 2 by 2 rectangle joined with no overlap have a total area of 14 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 67 • BACKWARD 2 • Q4/5

True or false?

A 3 by 3 rectangle and a 3 by 1 rectangle joined with no overlap have a total area of 10 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 67 • BACKWARD 2 • Q5/5

SET 67 • 7.G.B.6

FORWARD 1

Find the volume of a solid made of a prism and a cylinder • Grade 8 • Geometry • 8.G.C.9

A cylinder with a radius of 2 cm and a height of 6 cm stands on top of a 10 cm by 10 cm by 4 cm block.

Find the total volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 67 • FORWARD 1 • Q1/5

A cylinder with a radius of 3 inches and a height of 8 inches stands on top of a cube with 8-inch edges.

Find the total volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 67 • FORWARD 1 • Q2/5

A cube with 6 cm edges has a cylinder-shaped hole with a radius of 1 cm drilled straight through it (6 cm long).

What volume is left?

A. 216 + 6π cubic centimeters

B. 216 − 36π cubic centimeters

C. 6π cubic centimeters

D. 216 − 6π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 67 • FORWARD 1 • Q3/5

True or false?

A 2 by 2 by 10 prism with half of a cylinder (radius 1, length 10) on top has a volume of $40 + 5\pi$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 67 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 1 and a height of 5 is stacked on a cylinder with a radius of 1 and a height of 3. The total volume is 15π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 67 • FORWARD 1 • Q5/5

SET 67 • 7.G.B.6

FORWARD 2

Use volume formulas to solve problems with composite solids • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A silo is a cylinder with a radius of 3 m and a height of 10 m, topped with half of a sphere with a radius of 3 m.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 67 • FORWARD 2 • Q1/5

A pencil is a cylinder with a radius of 0.4 cm and a length of 15 cm, with a cone-shaped tip with a radius of 0.4 cm and a height of 1.5 cm.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 67 • FORWARD 2 • Q2/5

A cone with a radius of 3 and a height of 4 is carved out of a cylinder with a radius of 3 and a height of 4.

What volume is left?

A. 24π

B. 48π

C. 36π

D. 12π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q3**

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 67 • FORWARD 2 • Q3/5

True or false?

A square pyramid with 6 m base edges and a height of 4 m sits on top of a cube with 6 m edges. The total volume is 264 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 67 • FORWARD 2 • Q4/5

True or false?

Two halves of a sphere with a radius of 2 are joined to make a whole sphere. Its volume is $\frac{16}{3}\pi$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 67 • FORWARD 2 • Q5/5

SET 68 • 7.G.B.6 • GEOMETRY

Find the exterior surface area of a solid made of right prisms

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Find the surface area of a rectangular prism	**6.G.A.4**
BACKWARD 2	Find the area of a rectangle by multiplying	**3.MD.C.7.b**
FORWARD 1	Use the Pythagorean theorem to find a slanted edge needed for s…	**8.G.B.7**
FORWARD 2	Apply surface area to design problems	**HSG-MG.A.3**

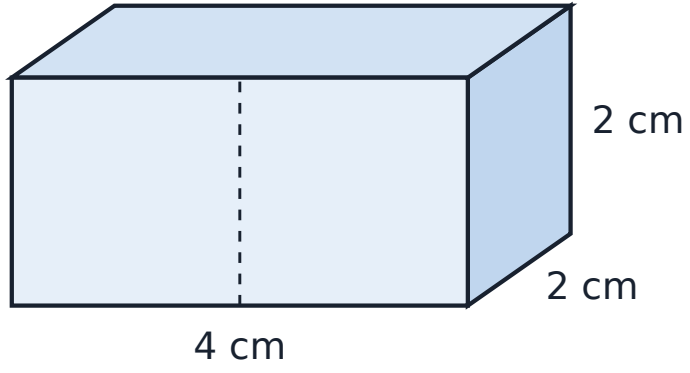
SET 68 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

Two cubes with 2 cm edges are joined face to face.

Find the exterior surface area of the solid. (The faces where the cubes touch are hidden and are not counted.)



Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 68 • MAIN • Q1/5

A cube with 2 m edges sits on top of a 6 m by 6 m by 2 m prism. Its bottom face lies completely on the top face of the prism.

Find the exterior surface area. (Do not count the faces where the solids touch.)

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 68 • MAIN • Q2/5

The solid is made of two right rectangular prisms joined face to face.

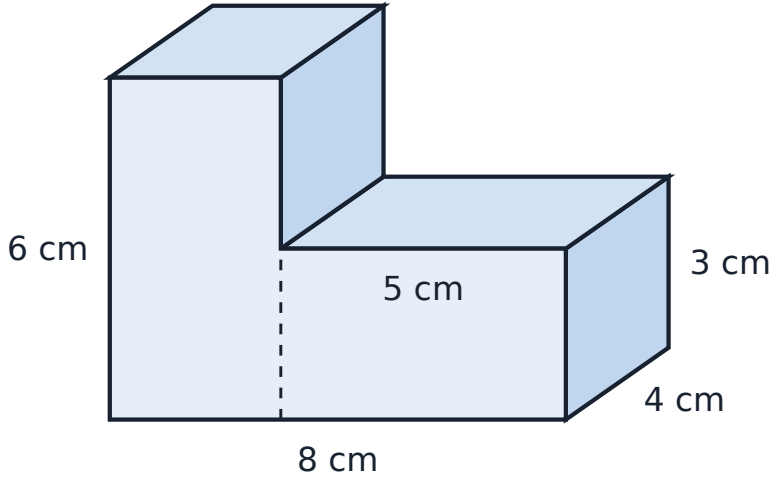
What is its exterior surface area?

A. 132 square centimeters

B. 178 square centimeters

C. 112 square centimeters

D. 210 square centimeters



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 68 • MAIN • Q3/5

True or false?

Two 1 by 1 by 1 cubes joined face to face have an exterior surface area of 10 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 68 • MAIN • Q4/5

True or false?

When two prisms are joined face to face, the exterior surface area is the sum of their two surface areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 68 • MAIN • Q5/5

SET 68 • 7.G.B.6

BACKWARD 1

Find the surface area of a rectangular prism • Grade 6 • Geometry • 6.G.A.4

Use a net to find the surface area of a 3 by 2 by 1 rectangular prism.

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 68 • BACKWARD 1 • Q1/5

Use a net to find the surface area of a cube with 3-inch edges.

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 68 • BACKWARD 1 • Q2/5

What is the surface area of a 4 by 1 by 1 rectangular prism?

A. 4 square units

B. 16 square units

C. 18 square units

D. 24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 68 • BACKWARD 1 • Q3/5

True or false?

A 2 by 2 by 1 rectangular prism has a surface area of 16 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 68 • BACKWARD 1 • Q4/5

True or false?

A 5 by 1 by 1 rectangular prism has a surface area of 20 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • 7.G.B.6

BACKWARD 2

Find the area of a rectangle by multiplying • Grade 3 • Measurement & Data • 3.MD.C.7.b

A face of a box is 6 cm by 4 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 68 • BACKWARD 2 • Q1/5

A face of a box is 7 m by 3 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 68 • BACKWARD 2 • Q2/5

A face of a box is 5 inches by 5 inches.

What is its area?

A. 20 square inches

B. 25 square inches

C. 30 square inches

D. 10 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 68 • BACKWARD 2 • Q3/5

True or false?

A 9 by 2 face has an area of 18 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 68 • BACKWARD 2 • Q4/5

True or false?

A 4 by 8 face has an area of 24 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 68 • BACKWARD 2 • Q5/5

SET 68 • 7.G.B.6

FORWARD 1

Use the Pythagorean theorem to find a slanted edge needed for surface area • Grade 8 • Geometry • 8.G.B.7

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A ramp is a right triangular prism. Its triangular ends have legs of 3 m and 4 m.

How wide is the sloped face (the hypotenuse of the end)?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 68 • FORWARD 1 • Q1/5 • NEAREST RELATED

A tent is a triangular prism. Each end is an isosceles triangle with a base of 6 feet and a height of 4 feet.

How long is each slanted side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 68 • FORWARD 1 • Q2/5 • NEAREST RELATED

The end of a prism is a right triangle with legs of 5 and 12.

How long is its hypotenuse?

A. 13

B. 17

C. 7

D. $\sqrt{119}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 68 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A ramp with triangular ends whose legs are 6 and 8 has a sloped face that is 10 units wide.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 68 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

A tent end is an isosceles triangle with a base of 10 and a height of 12. Each slanted side is 12 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 68 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 68 • 7.G.B.6

FORWARD 2

Apply surface area to design problems • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A gift is made of two cubes with 3-inch edges joined face to face.

How much wrapping paper covers its outside exactly?

Paper: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 68 • FORWARD 2 • Q1/5

A shed is a 10 ft by 8 ft by 8 ft rectangular prism (8 ft tall). Its four walls and its flat roof are painted, but not the floor. One quart of paint covers 40 square feet.

How many whole quarts are needed?

Quarts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 68 • FORWARD 2 • Q2/5

Two 1 by 2 by 3 prisms are joined face to face with matching faces.

Which way of joining them gives the least exterior surface area?

A. Joining their 1 by 2 faces

B. Joining their 1 by 3 faces

C. Joining their 2 by 3 faces

D. All three ways give the same surface area

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q3**

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 68 • FORWARD 2 • Q3/5

True or false?

Joining two solids face to face lowers the total exterior surface area by twice the area of the shared face.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 68 • FORWARD 2 • Q4/5

True or false?

A box with a smaller surface area always has a smaller volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 68 • FORWARD 2 • Q5/5

SET 49 • 7.SP.A.1 • STATISTICS & PROBABILITY

Understand random sampling and representative samples

MAIN	Grade 7 main question	**7.SP.A.1**
BACKWARD 1	Recognize a statistical question	**6.SP.A.1**
BACKWARD 2	Report the number of observations	**6.SP.B.5.a**
FORWARD 1	Summarize survey data in a two-way table — nearest related (no…	**8.SP.A.4**
FORWARD 2	Recognize surveys, experiments, and observational studies	**HSS-IC.B.3**

SET 49 • 7.SP.A.1

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.1

A principal wants to know students' favorite lunch.

Which sample is most likely to represent all the students in the school?

A. 20 students in one 7th grade class

B. 20 students on the basketball team

C. The first 20 students in the lunch line

D. 20 students chosen at random from the school list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q1/5

A town surveys people at a gym about how often they exercise.

Is the sample likely to represent the whole town? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q2/5

Describe one way to choose a random sample of 30 students from a school of 600 students.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q3/5

True or false?

In a random sample, every member of the population has the same chance of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q4/5

True or false?

A large sample always represents the population, even if it is not chosen at random.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q5/5

SET 49 • 7.SP.A.1

BACKWARD 1

Recognize a statistical question • Grade 6 • Statistics & Probability • 6.SP.A.1

Is "How tall are the students in my class?" a statistical question? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q1/5

Which is a statistical question?

A. How many hours do 7th graders sleep on school nights?

B. What is 5 × 8?

C. How tall is the school flagpole?

D. What day is it today?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q2/5

Rewrite "How old am I?" as a statistical question.

Question: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q3/5

True or false?

"What are the shoe sizes of the players on the team?" is a statistical question.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q4/5

True or false?

"How many days are in March?" is a statistical question.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • 7.SP.A.1

BACKWARD 2

Report the number of observations • Grade 6 • Statistics & Probability • 6.SP.B.5.a

How many observations does the dot plot show?



Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q1/5

A survey asks 25 students how many pets they have. Each student gives one answer.

How many observations are there?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q2/5

A table lists the heights of 18 plants, one height for each plant.

How many observations are there?

A. 1

B. 36

C. 18

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q3/5

True or false?

Each dot on a dot plot shows one observation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q4/5

True or false?

A data set of 12 test scores has 24 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q5/5

SET 49 • 7.SP.A.1

FORWARD 1

Summarize survey data in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows the results of a survey.

What fraction of the surveyed students play a sport?

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows the results of a survey.

What fraction of all surveyed students have a job and play a sport?

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows the results of a survey.

Among students who have a job, what fraction play a sport?

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

A. $\frac{3}{10}$ **B.** $\frac{3}{5}$ **C.** $\frac{2}{5}$ **D.** $\frac{3}{20}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, 56% of the students with no job play a sport.

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, more than half of all the students have a job.

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 49 • 7.SP.A.1

FORWARD 2

Recognize surveys, experiments, and observational studies • High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers randomly assign 50 plants to get a new plant food and 50 plants to get none.

Is this a survey, an experiment, or an observational study?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q1/5

A study records the screen time of 200 teens without changing anything they do.

Is this a survey, an experiment, or an observational study?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q2/5

Why is random assignment used in an experiment?

A. To avoid collecting data

B. To make the sample larger

C. To make the groups alike, so differences can be linked to the treatment

D. To let people choose their own group

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q3/5

True or false?

A survey asks a sample of people questions in order to estimate facts about a population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q4/5

True or false?

An observational study can prove that one variable causes another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q5/5

SET 50 • 7.SP.A.2 • STATISTICS & PROBABILITY

Use data from a random sample to draw inferences about a population

MAIN	Grade 7 main question	**7.SP.A.2**
BACKWARD 1	Find what percent a part is of a whole	**6.RP.A.3.c**
BACKWARD 2	Find a missing value in equivalent ratios	**6.RP.A.3.a**
FORWARD 1	Use a linear model to make a prediction — nearest related (no di…	**8.SP.A.3**
FORWARD 2	Use a margin of error to describe a population estimate	**HSS-IC.B.4**

SET 50 • 7.SP.A.2

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.2

In a random sample of 50 students, 18 prefer the art club.

Predict how many of the 800 students in the school prefer the art club.

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q1/5

In a random sample of 40 light bulbs, 2 do not work.

Estimate how many bulbs in a shipment of 2,000 do not work.

Bulbs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q2/5

In a random sample of 60 voters, 33 support a plan.

What is the best estimate of the percent of all voters who support it?

A. 45%

B. 33%

C. 60%

D. 55%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q3/5

True or false?

In a random sample of 25 fish from a lake, 5 are trout. A good estimate is that 20% of the fish in the lake are trout.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q4/5

True or false?

In a random sample of 30 homes, 9 have a dog. A good estimate is that 900 of 1,500 homes have a dog.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q5/5

SET 50 • 7.SP.A.2

BACKWARD 1

Find what percent a part is of a whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

18 out of 50 is what percent?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q1/5

10 is what percent of 40?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q2/5

7 out of 20 is what percent?

A. 35%

B. 7%

C. 20%

D. 70%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q3/5

True or false?

12 out of 48 is 25%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q4/5

True or false?

9 out of 30 is 9%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 7.SP.A.2

BACKWARD 2

Find a missing value in equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Find the missing value.

$2 : 5 = ? : 100$

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q1/5

Find the missing value.

$4 : 9 = 28 : ?$

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q2/5

Which ratio is equivalent to 6 : 15?

A. 3 : 5

B. 6 : 5

C. 2 : 5

D. 1 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q3/5

True or false?

$5 : 8 = 25 : 40$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q4/5

True or false?

$3 : 7 = 9 : 14$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 7.SP.A.2

FORWARD 1

Use a linear model to make a prediction • Grade 8 • Statistics & Probability • 8.SP.A.3

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A line of fit is y = 2.5x + 40, where x is hours studied and y is a test score.

Predict the score for 8 hours of study.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q1/5 • NEAREST RELATED

A line of fit is $y = 2.5x + 40$, where x is hours studied and y is a test score.

What does 2.5 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q2/5 • NEAREST RELATED

A line of fit is $y = -3x + 90$.

What does it predict for $x = 10$?

A. 120

B. 93

C. 87

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

For the line of fit $y = 0.5x + 12$, the predicted y-value at $x = 0$ is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

The line of fit $y = 4x + 1$ predicts $y = 41$ when $x = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 50 • 7.SP.A.2

FORWARD 2

Use a margin of error to describe a population estimate • High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample shows 52% support for a plan, with a margin of error of 4%.

What interval of values is likely for the whole population?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q1/5

A random sample has a mean of 6.8 hours with a margin of error of 0.5 hour.

What interval of values is likely for the population mean?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q2/5

As a random sample gets larger, what usually happens to the margin of error?

A. It gets larger.

B. It becomes 0 once the sample is over 10.

C. It gets smaller.

D. It stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q3/5

True or false?

A margin of error describes how far a sample estimate is likely to be from the population value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q4/5

True or false?

A poll result of 52% with a 4% margin of error proves that more than half of the population supports the plan.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q5/5

SET 51 • 7.SP.A.2 • STATISTICS & PROBABILITY

Use several random samples to judge how much estimates vary

MAIN	Grade 7 main question	**7.SP.A.2**
BACKWARD 1	Find the mean of a data set	**6.SP.B.5.c**
BACKWARD 2	Find the range as a measure of variation	**6.SP.A.3**
FORWARD 1	Describe the association shown in a scatter plot — nearest relate…	**8.SP.A.1**
FORWARD 2	Decide whether a result is consistent with a model by simulation	**HSS-IC.A.2**

SET 51 • 7.SP.A.2

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.2

Four random samples of 20 students give these mean hours of sleep: 7.8, 8.2, 8.0, and 8.4.

What is a good estimate of the mean for all the students? Use the mean of the samples.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q1/5

Four random samples give mean estimates of 7.8, 8.2, 8.0, and 8.4 hours.

What is the range of the estimates?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q2/5

Three random samples of 50 people show that 22%, 26%, and 24% prefer blue.

Which conclusion is best?

A. Exactly 22% of the population prefers blue.

B. Exactly 26% of the population prefers blue.

C. The samples are useless because they differ.

D. About 24% of the population prefers blue.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q3/5

True or false?

Different random samples from the same population usually give somewhat different results.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q4/5

True or false?

Small random samples usually give estimates that vary less than large random samples.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q5/5

SET 51 • 7.SP.A.2

BACKWARD 1

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

4, 7, 9, 10

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q1/5

Find the mean.

12, 15, 18

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q2/5

What is the mean of 3, 3, 6, and 8?

A. 5

B. 6

C. 3

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q3/5

True or false?

The mean of 2, 4, 6, and 8 is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q4/5

True or false?

The mean of 10, 20, and 60 is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 7.SP.A.2

BACKWARD 2

Find the range as a measure of variation • Grade 6 • Statistics & Probability • 6.SP.A.3

Find the range.

12, 18, 9, 15

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q1/5

Find the range.

3.5, 7, 4.25, 6

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q2/5

Which data set has the greatest range?

A. 1, 3, 2

B. 7, 9, 8

C. 4, 6, 5

D. 2, 10, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q3/5

True or false?

The range of 20, 25, and 30 is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q4/5

True or false?

A data set whose values are all the same has a range of 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • 7.SP.A.2

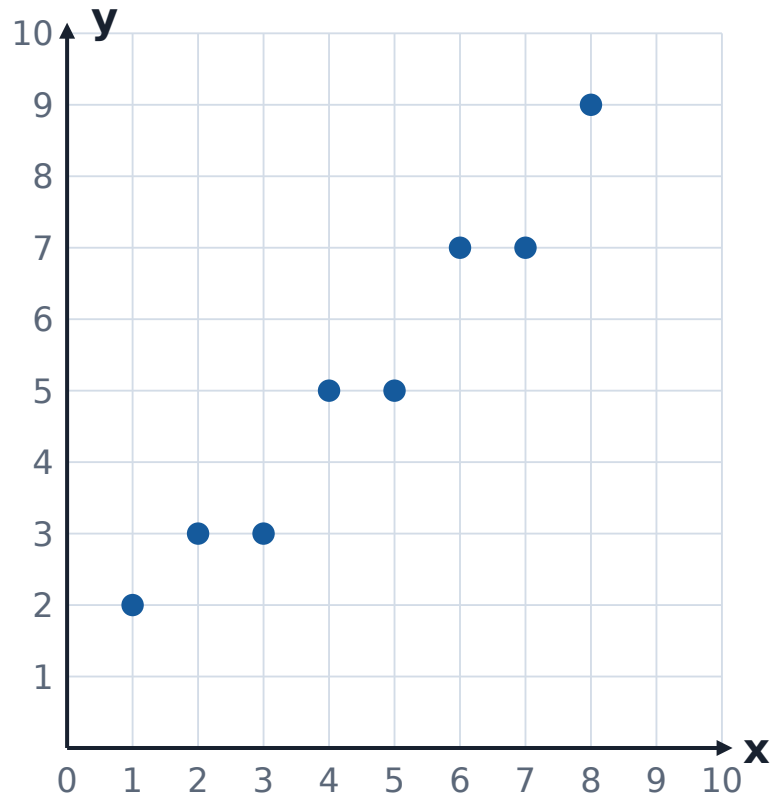
FORWARD 1

Describe the association shown in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

What kind of association does the scatter plot show?

Association: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q1/5 • NEAREST RELATED

A scatter plot shows hours of practice (x) and free throws made (y). The points rise from left to right in a roughly straight pattern.

Describe the association.

Association: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q2/5 • NEAREST RELATED

A scatter plot shows the age of a car (x) and its value (y).

What association is most likely?

A. Negative association

B. Every point on one line

C. No association

D. Positive association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In a scatter plot with no association, the points show no clear trend.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

Points that fall from left to right in a scatter plot show a positive association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 51 • 7.SP.A.2

FORWARD 2

Decide whether a result is consistent with a model by simulation • High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A coin is said to be fair. In 100 simulated sets of 20 fair flips, 15 or more heads happened only 2 times. A real coin gives 15 heads in 20 flips.

Is the fair-coin model believable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q1/5

Simulations of a spinner model usually give 3 to 7 reds in 20 spins. A real spinner gives 5 reds in 20 spins.

What does this show?

A. The spinner is proven unfair.

B. The result is impossible.

C. The result is consistent with the model.

D. The model is proven wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q2/5

In 100 simulations of a model, a certain result happened 40 times.

Is that result unusual for the model? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q3/5

True or false?

A result that rarely happens in simulations of a model suggests that the model may be wrong.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q4/5

True or false?

A result that happens even once in 1,000 simulations is a common result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q5/5

SET 69 • 7.SP.A.2 • STATISTICS & PROBABILITY

Generate several same-size random samples with random digits and see how the estimates vary

MAIN	Grade 7 main question	**7.SP.A.2**
BACKWARD 1	Write a part of a small sample as a percent	**6.RP.A.3.c**
BACKWARD 2	Find the mean of a small data set	**6.SP.B.5.c**
FORWARD 1	Summarize categorical data from two random samples in a two-…	**8.SP.A.4**
FORWARD 2	Use the spread of simulated samples to give an interval for a popu…	**HSS-IC.B.4**

SET 69 • 7.SP.A.2

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.2

A school has 100 students numbered 00 to 99. Students 00 to 39 chose art as their favorite class.

Use these random two-digit numbers to choose a sample of 5 students:

07 52 31 88 14

What percent of the sample chose art?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 69 • MAIN • Q1/5

A school has 100 students numbered 00 to 99. Students 00 to 39 chose art as their favorite class.

Use these random two-digit numbers to choose a sample of 5 students:

45 63 02 97 71

What percent of the sample chose art?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 69 • MAIN • Q2/5

Students 00 to 39 of a school of 100 chose art. Four random samples of 5 students were chosen with random digits:

Sample 1: 07 52 31 88 14

Sample 2: 45 63 02 97 71

Sample 3: 38 19 60 25 84

Sample 4: 50 11 76 93 66

What is the range of the four sample percents?

A. 60 percentage points

B. 0 percentage points

C. 20 percentage points

D. 40 percentage points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 69 • MAIN • Q3/5

Students 00 to 39 of a school of 100 chose art, so the true percent is 40%. Four random samples of 5 students gave 60%, 20%, 60%, and 20%.

What is the mean of the four sample percents? How does it compare with the true percent, and how much do single samples vary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 69 • MAIN • Q4/5

True or false?

Random samples of 5 students gave estimates from 20% to 60%. Random samples of 50 students from the same school would usually give estimates that are closer together.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 69 • MAIN • Q5/5

SET 69 • 7.SP.A.2

BACKWARD 1

Write a part of a small sample as a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write 3 out of 5 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 69 • BACKWARD 1 • Q1/5

Write 2 out of 5 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 69 • BACKWARD 1 • Q2/5

What percent is 4 out of 5?

A. 80%

B. 20%

C. 45%

D. 4%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 69 • BACKWARD 1 • Q3/5

True or false?

1 out of 5 is 20%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 69 • BACKWARD 1 • Q4/5

True or false?

3 out of 5 is 35%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 69 • BACKWARD 1 • Q5/5

SET 69 • 7.SP.A.2

BACKWARD 2

Find the mean of a small data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

60, 20, 60, 20

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 69 • BACKWARD 2 • Q1/5

Find the mean.

30, 50, 40

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 69 • BACKWARD 2 • Q2/5

What is the mean of 20, 40, 40, and 60?

A. 50

B. 35

C. 45

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 69 • BACKWARD 2 • Q3/5

True or false?

The mean of 10, 30, and 50 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 69 • BACKWARD 2 • Q4/5

True or false?

The mean of 20, 20, and 80 is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 69 • BACKWARD 2 • Q5/5

SET 69 • 7.SP.A.2

FORWARD 1

Summarize categorical data from two random samples in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows random samples of 30 students from each grade.

What percent of the Grade 7 sample chose art?

	Chose art	Did not	Total
Grade 7 sample	12	18	30
Grade 8 sample	9	21	30
Total	21	39	60

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 69 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows random samples of 30 students from each grade.

What percent of the Grade 8 sample chose art?

	Chose art	Did not	Total
Grade 7 sample	12	18	30
Grade 8 sample	9	21	30
Total	21	39	60

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 69 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows random samples of 30 students from each grade.

Which conclusion do the samples best support?

	Chose art	Did not	Total
Grade 7 sample	12	18	30
Grade 8 sample	9	21	30
Total	21	39	60

A. Every Grade 7 student prefers art.

B. A greater share of Grade 7 students than Grade 8 students may prefer art.

C. No Grade 8 student prefers art.

D. Exactly 40% of all Grade 7 students prefer art.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 69 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of choosing art in the whole sample of 60 students is 0.35.

	Chose art	Did not	Total
Grade 7 sample	12	18	30
Grade 8 sample	9	21	30
Total	21	39	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 69 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, more Grade 8 students than Grade 7 students chose art.

	Chose art	Did not	Total
Grade 7 sample	12	18	30
Grade 8 sample	9	21	30
Total	21	39	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 69 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 69 • 7.SP.A.2

FORWARD 2

Use the spread of simulated samples to give an interval for a population percent • High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

Simulations show that the middle 95% of random samples of 50 students land within 10 percentage points of the true percent. A real random sample of 50 gives 44%.

What interval is likely to contain the population percent?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 69 • FORWARD 2 • Q1/5

Simulations show that the middle 95% of random samples of 200 people land within 7 percentage points of the true percent. A real random sample of 200 gives 58%.

What interval is likely to contain the population percent?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 69 • FORWARD 2 • Q2/5

In simulations, how does the spread of sample percents change as the samples get larger?

A. It gets larger.

B. It stays the same.

C. It gets smaller.

D. It becomes 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 69 • FORWARD 2 • Q3/5

True or false?

An interval of 34% to 54% means the population percent is likely between those values, but it is not certain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 69 • FORWARD 2 • Q4/5

True or false?

Simulated samples of 50 vary by about 10 percentage points. So one real sample of 50 always lands exactly on the population percent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 69 • FORWARD 2 • Q5/5

SET 52 • 7.SP.B.3 • STATISTICS & PROBABILITY

Describe how much two distributions overlap using the difference in means as a multiple of the MAD

MAIN	Grade 7 main question	**7.SP.B.3**
BACKWARD 1	Find the mean absolute deviation (MAD)	**6.SP.B.5.c**
BACKWARD 2	Describe the shape of a distribution	**6.SP.A.2**
FORWARD 1	Compare groups using relative frequencies in a two-way table —…	**8.SP.A.4**
FORWARD 2	Compare the center and spread of two data sets	**HSS-ID.A.2**

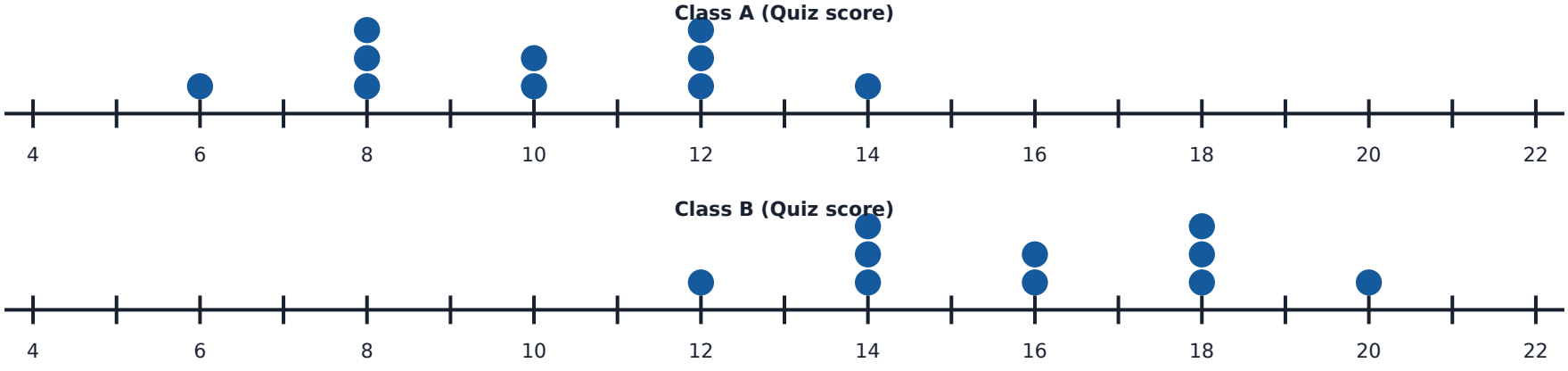
SET 52 • 7.SP.B.3

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.B.3

The dot plots show quiz scores for two classes. The means are 10 and 16, and each data set has a MAD of 2.

Write the difference in the means as a multiple of the MAD.



Answer: _____

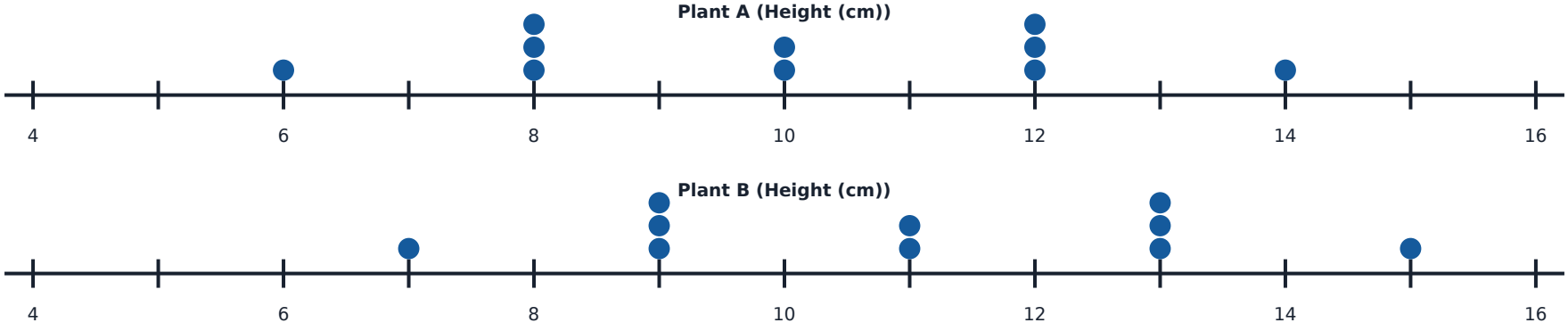
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q1/5

The dot plots show the heights of two kinds of plants, in centimeters. The means are 10 and 11, and each data set has a MAD of 2.

Write the difference in the means as a multiple of the MAD, and describe how much the dot plots overlap.



Answer: _____

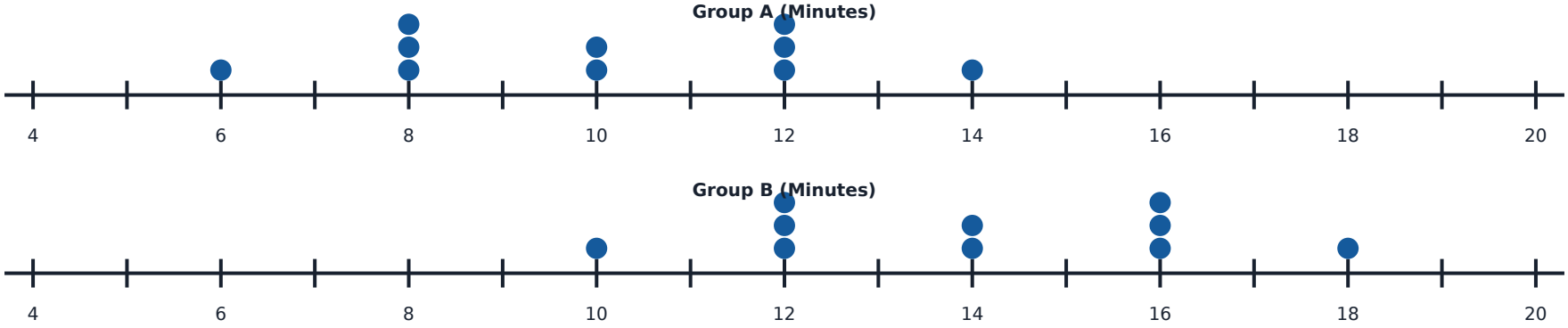
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q2/5

The dot plots show the minutes two groups spent reading. The means are 10 and 14, and each data set has a MAD of 2.

Which statement describes the dot plots?



A. The means differ by 2 MADs, and the plots overlap somewhat.

B. The means are the same, so the plots overlap completely.

C. The means differ by 2 MADs, so the plots cannot share any values.

D. The means differ by 4 MADs, and the plots do not overlap.

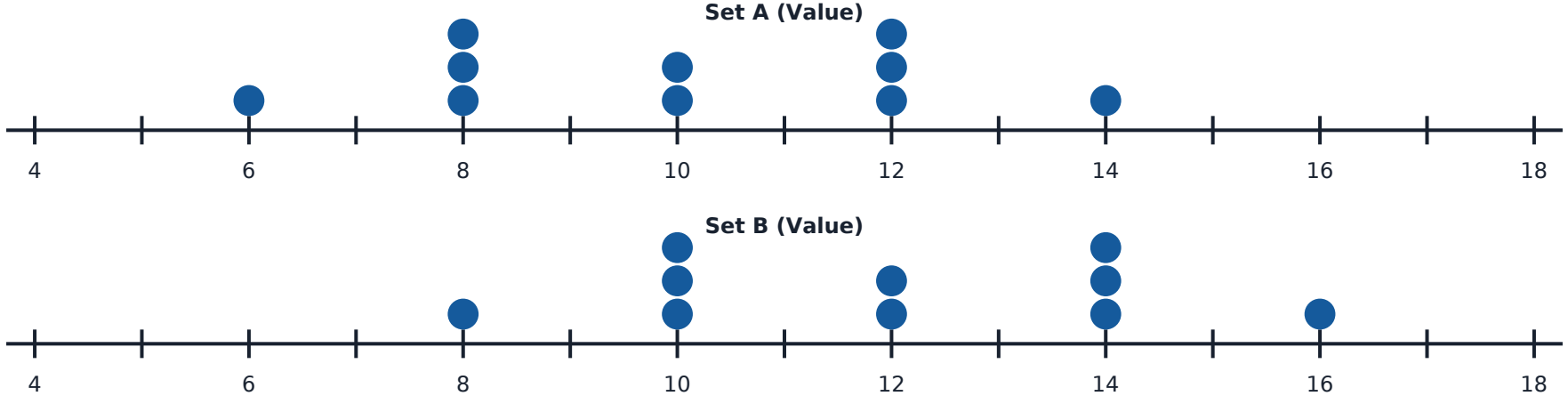
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q3/5

True or false?

The dot plots show two data sets with means of 10 and 12, and each has a MAD of 2. The means differ by 1 MAD, and the dot plots overlap a lot.



A. True

B. False

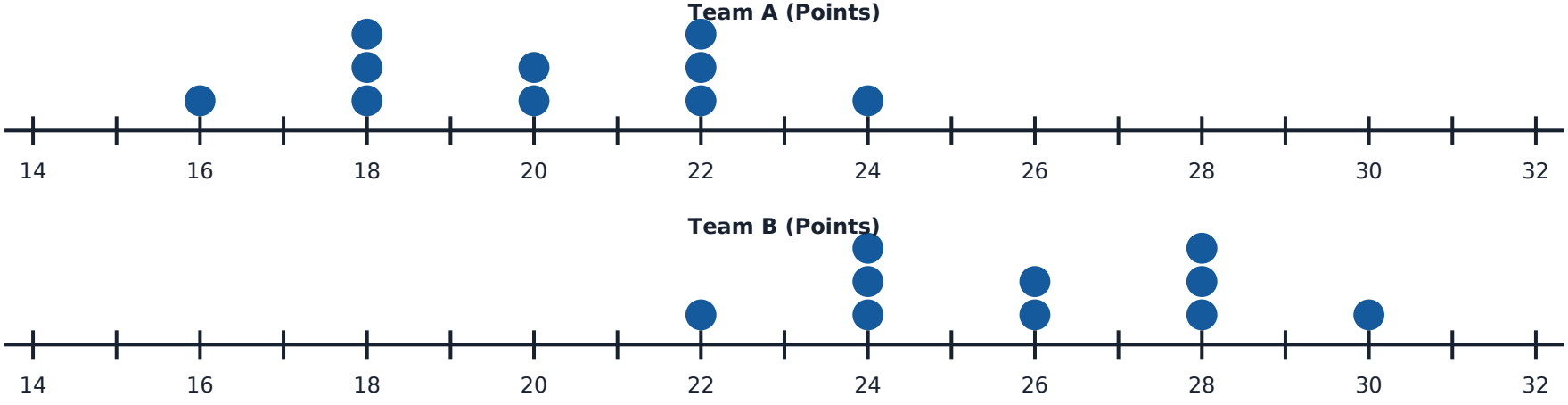
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q4/5

True or false?

The dot plots show two data sets with means of 20 and 26, and each has a MAD of 2. The means differ by 3 MADs, and the two dot plots share no values.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q5/5

SET 52 • 7.SP.B.3

BACKWARD 1

Find the mean absolute deviation (MAD) • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the MAD.

2, 4, 6, 8

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q1/5

Find the MAD.

5, 5, 9, 13

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q2/5

What is the MAD of 10, 10, 10, 10?

A. 0

B. 40

C. 1

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q3/5

True or false?

The MAD of 1, 3, 5, 7 is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q4/5

True or false?

The MAD of 0 and 10 is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 7.SP.B.3

BACKWARD 2

Describe the shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

A dot plot has most of its dots on the left and a long tail to the right.

How is its shape described?

A. Uniform

B. Skewed right

C. Skewed left

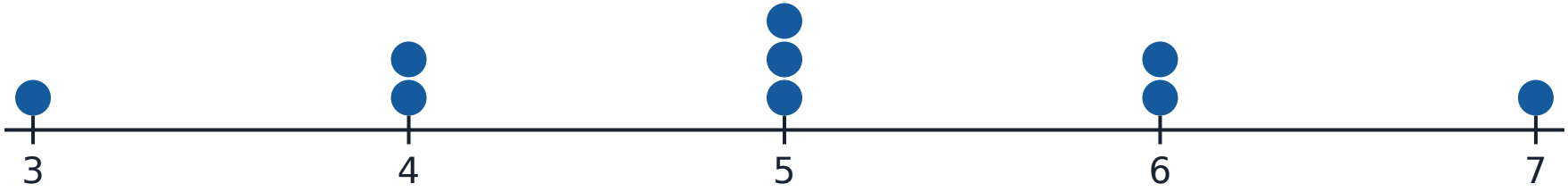
D. Symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q1/5

Is the distribution in the dot plot symmetric?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q2/5

A dot plot has a tall stack of dots in the middle and short, equal stacks on each side.

What is the shape of the distribution?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q3/5

True or false?

A distribution that looks the same on both sides of its center is symmetric.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q4/5

True or false?

A dot plot with all of its dots stacked above one value is skewed left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • 7.SP.B.3

FORWARD 1

Compare groups using relative frequencies in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows how students get to school.

What percent of sixth graders walk?

	Walks	Rides	Total
Sixth grade	24	36	60
Seventh grade	15	45	60
Total	39	81	120

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows how students get to school.

What percent of seventh graders walk?

	Walks	Rides	Total
Sixth grade	24	36	60
Seventh grade	15	45	60
Total	39	81	120

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows how students get to school.

Is there evidence that grade and walking are related?

	Walks	Rides	Total
Sixth grade	24	36	60
Seventh grade	15	45	60
Total	39	81	120

A. No, because each grade has 60 students.

B. No, because more students ride than walk in both grades.

C. Yes, because 40% of sixth graders walk but only 25% of seventh graders do.

D. Yes, because 24 is greater than 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

Relative frequencies make it possible to compare groups of different sizes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

If 40% of sixth graders and 40% of seventh graders walk, there is a strong relationship between grade and walking.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 52 • 7.SP.B.3

FORWARD 2

Compare the center and spread of two data sets • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Data set A has a mean of 20 and a standard deviation of 2. Data set B has a mean of 20 and a standard deviation of 6.

Which data set is more spread out?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q1/5

Data set A has a median of 45 and an IQR of 10. Data set B has a median of 52 and an IQR of 10.

Compare the centers and the spreads.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q2/5

Which measure of spread is best for a skewed data set with outliers?

A. Interquartile range

B. Mean

C. Range

D. Standard deviation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q3/5

True or false?

Standard deviation measures how spread out the data are around the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q4/5

True or false?

Two data sets with the same mean must have the same standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q5/5

SET 53 • 7.SP.B.4 • STATISTICS & PROBABILITY

Compare two populations using measures of center and variability from random samples

MAIN	Grade 7 main question	**7.SP.B.4**
BACKWARD 1	Find the interquartile range	**6.SP.B.5.c**
BACKWARD 2	Read a box plot	**6.SP.B.4**
FORWARD 1	Identify clusters and outliers in a scatter plot — nearest related (n…	**8.SP.A.1**
FORWARD 2	Interpret differences in center and spread, including the effect of…	**HSS-ID.A.3**

SET 53 • 7.SP.B.4

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.B.4

Random samples of quiz scores: Class A has a median of 78 and an IQR of 10. Class B has a median of 70 and an IQR of 10.

What do the samples suggest about the two classes? Use both the center and the spread.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q1/5

A random sample of 7th graders has a mean of 6.5 hours of screen time and a MAD of 1.2. A random sample of 8th graders has a mean of 7.7 hours and a MAD of 1.2.

Write the difference in means as a multiple of the MAD. What does this suggest about the two populations?

Answer: _____

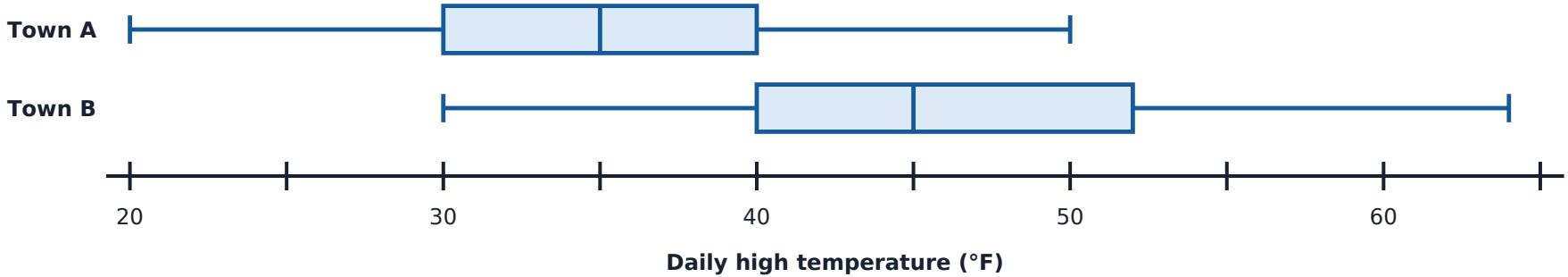
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q2/5

The box plots show random samples of daily high temperatures in two towns.

Which statement is true?



A. The two towns have the same range.

B. The IQRs differ by 10 degrees.

C. The median for Town B is 10 degrees greater than the median for Town A.

D. Town A has the greater median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q3/5

True or false?

Random samples from two populations have means of 20 and 30, and each has a MAD of 2. The samples suggest that values in the second population tend to be greater.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q4/5

True or false?

A random sample from population A has a mean of 52 and one from population B has a mean of 48, each with a MAD of 10. The samples prove that every member of population A has a greater value than every member of population B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q5/5

SET 53 • 7.SP.B.4

BACKWARD 1

Find the interquartile range • Grade 6 • Statistics & Probability • 6.SP.B.5.c

A data set has Q1 = 12 and Q3 = 20.

What is the IQR?

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q1/5

Find the IQR.

2, 4, 6, 8, 10, 12, 14, 16

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q2/5

A box plot has Q1 = 30 and Q3 = 55.

What is the IQR?

A. 30

B. 85

C. 25

D. 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q3/5

True or false?

The IQR is the difference between Q3 and Q1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q4/5

True or false?

A data set with Q1 = 7 and Q3 = 15 has an IQR of 22.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

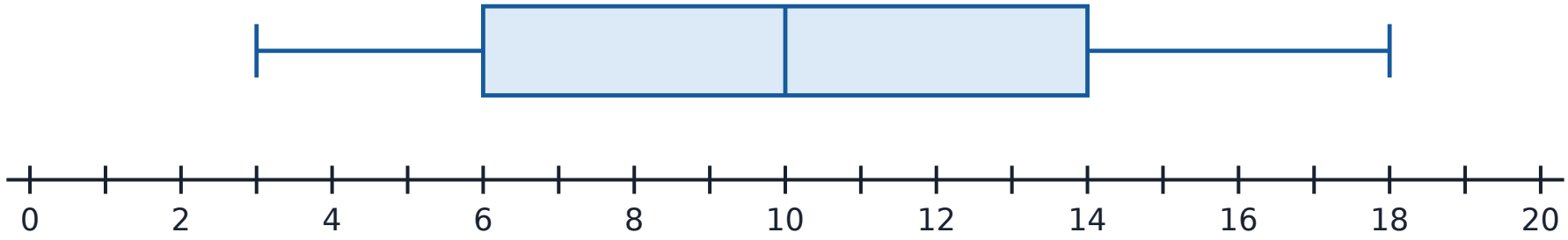
Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q5/5

SET 53 • 7.SP.B.4

BACKWARD 2

Read a box plot • Grade 6 • Statistics & Probability • 6.SP.B.4

What is the median of the data in the box plot?



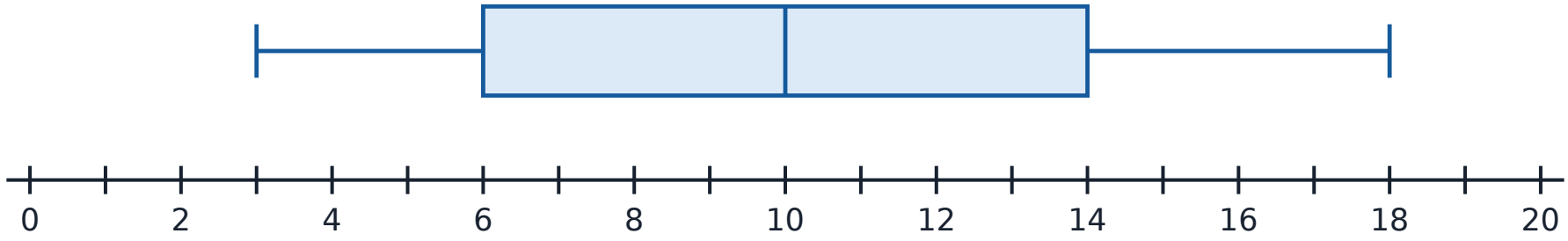
Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

What is the first quartile (Q1) of the data in the box plot?



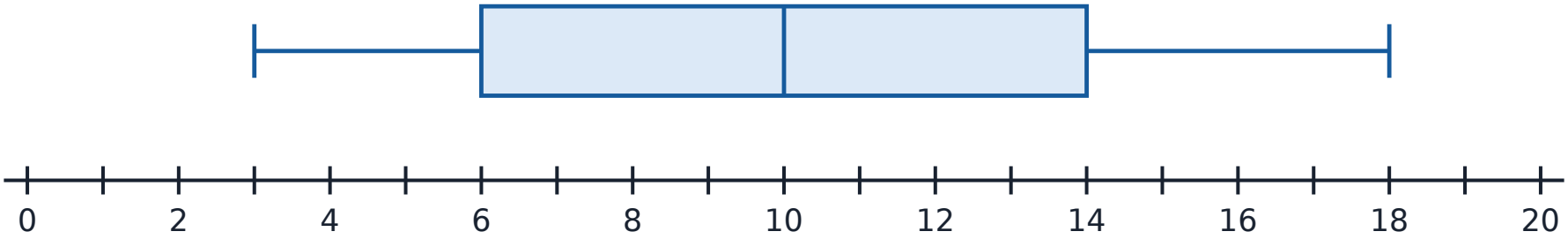
Q1: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

What is the maximum of the data in the box plot?



A. 14

B. 18

C. 3

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

True or false?

The box of a box plot shows the middle half of the data.

A. True

B. False

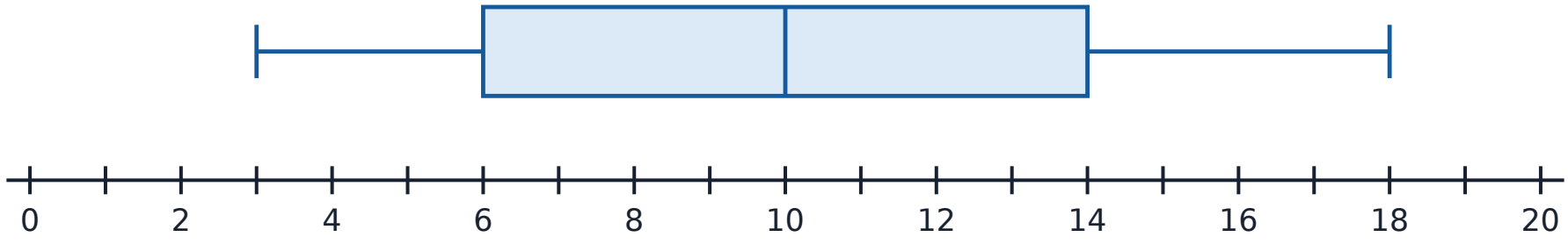
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

True or false?

The minimum of the data in the box plot is 6.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 7.SP.B.4

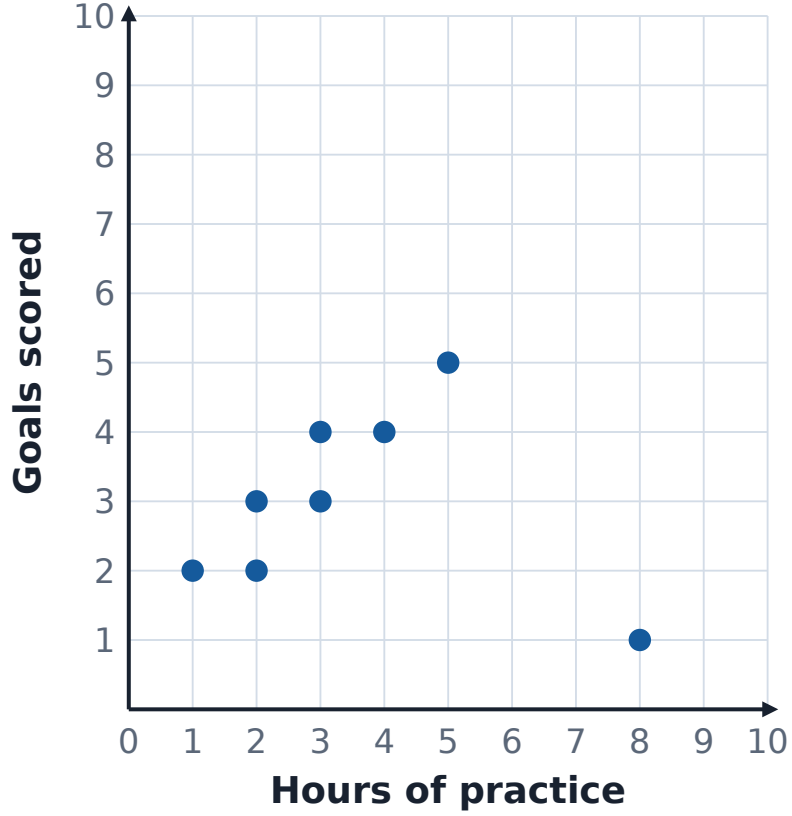
FORWARD 1

Identify clusters and outliers in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Which point in the scatter plot is an outlier?

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q1/5 • NEAREST RELATED

A scatter plot has a group of points close together.

What is the group called?

A. A cluster

B. An outlier

C. An intercept

D. A slope

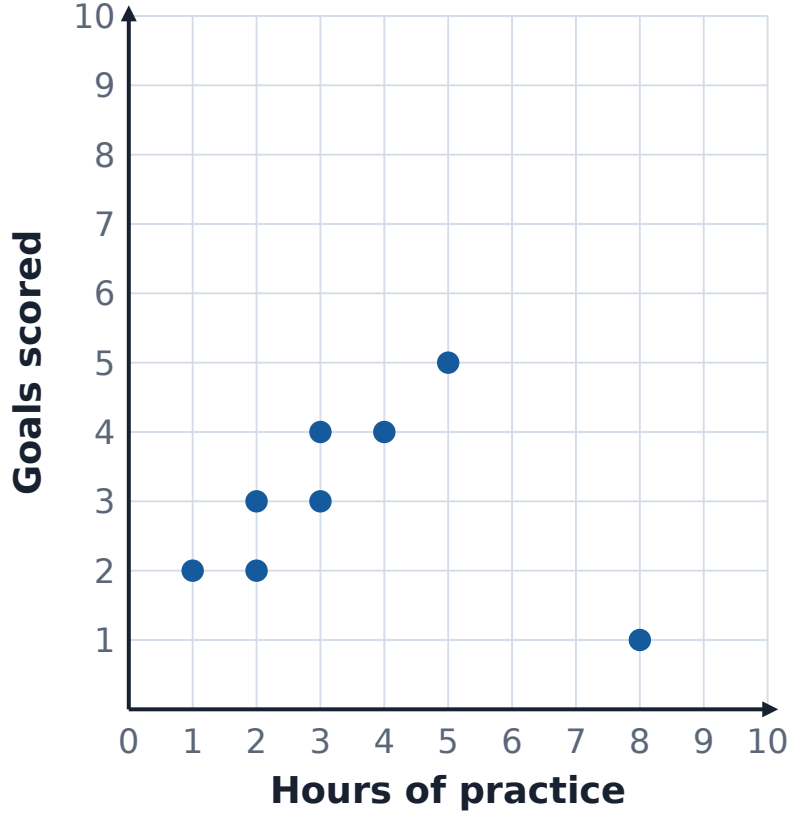
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q2/5 • NEAREST RELATED

Leaving out the outlier, describe the association in the scatter plot.

Association: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

An outlier is a point that is far from the overall pattern of the data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

Every scatter plot has exactly one outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 53 • 7.SP.B.4

FORWARD 2

Interpret differences in center and spread, including the effect of outliers • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

A data set is 4, 5, 5, 6, 7, 40.

Which better describes its center, the mean or the median? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q1/5

A data set is 4, 5, 5, 6, 7, 40.

Find the mean with the outlier and the mean without it. Round a mean that does not end to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q2/5

Which measure is affected most by an outlier?

A. Mean

B. IQR

C. Mode

D. Median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q3/5

True or false?

The median is resistant to outliers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q4/5

True or false?

Removing a very large outlier makes the mean larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q5/5

SET 54 • 7.SP.C.5 • STATISTICS & PROBABILITY

Understand probability as a number from 0 to 1 that shows how likely an event is

MAIN	Grade 7 main question	**7.SP.C.5**
BACKWARD 1	Compare two fractions	**4.NF.A.2**
BACKWARD 2	Place fractions on a number line from 0 to 1	**3.NF.A.2**
FORWARD 1	Find relative frequencies in a two-way table — nearest related (no…	**8.SP.A.4**
FORWARD 2	Describe unions, intersections, and complements of events as sub…	**HSS-CP.A.1**

SET 54 • 7.SP.C.5

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.5

Which probability describes an event that is likely to happen?

A. 0.1

B. 0.8

C. 0.5

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q1/5

A bag has 5 red marbles and no other marbles.

What is the probability of drawing a red marble?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

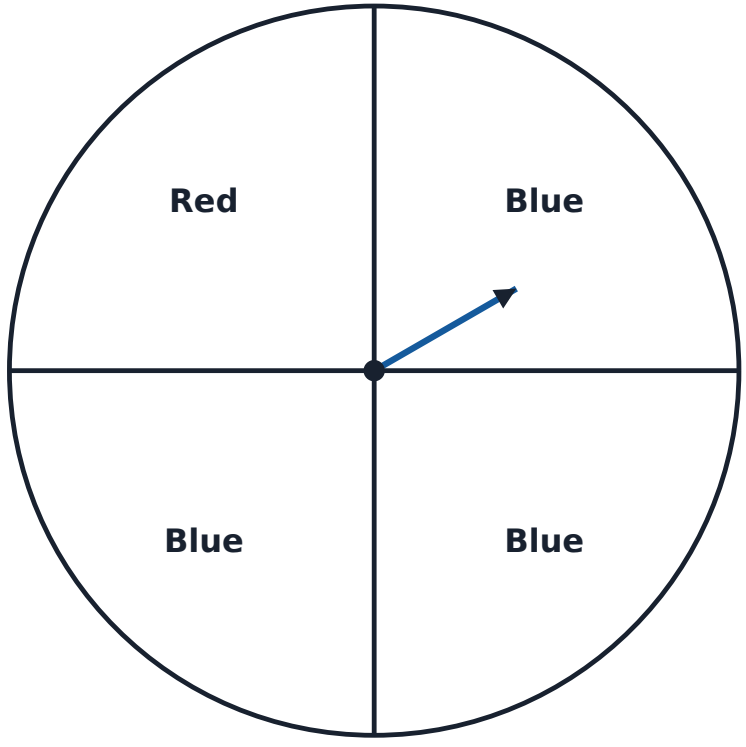
ID S54-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q2/5

The spinner has 4 equal sections.

Is landing on Blue likely, unlikely, or neither likely nor unlikely?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q3/5

True or false?

An event with a probability of $\frac{1}{2}$ is just as likely to happen as not to happen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q4/5

True or false?

A probability of 1.3 means that an event is very likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q5/5

SET 54 • 7.SP.C.5

BACKWARD 1

Compare two fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which is greater, $\frac{3}{8}$ or $\frac{2}{3}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q1/5

Write >, <, or =.

$\frac{5}{10}$ ____ $\frac{1}{2}$

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q2/5

Which fraction is less than $\frac{1}{2}$?

A. $\frac{3}{4}$

B. $\frac{5}{8}$

C. $\frac{3}{8}$

D. $\frac{4}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q3/5

True or false?

$\frac{7}{12} > \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q4/5

True or false?

$\frac{2}{5} > \frac{3}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • 7.SP.C.5

BACKWARD 2

Place fractions on a number line from 0 to 1 • Grade 3 • Number & Operations–Fractions • 3.NF.A.2

Plot $\frac{3}{4}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q1/5

A number line from 0 to 1 is split into 4 equal parts.

What fraction is at the first tick mark after 0?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q2/5

Which fraction is halfway between 0 and 1 on a number line?

A. $\frac{1}{2}$

B. $\frac{1}{4}$

C. $\frac{2}{1}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q3/5

True or false?

$\frac{4}{4}$ is at the same point as 1 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3}$ is farther from 0 than $\frac{2}{3}$ is.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 7.SP.C.5

FORWARD 1

Find relative frequencies in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows the results of a survey.

Among Grade 9 students, what is the relative frequency of playing music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows the results of a survey.

Among Grade 10 students, what is the relative frequency of playing music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows the results of a survey.

What is the relative frequency of all students who play music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. 0.3 **B.** 0.18 **C.** 0.12 **D.** 0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, students in both grades play music at the same rate.

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, Grade 9 students are more likely than Grade 10 students to play music, because 18 > 12.

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 54 • 7.SP.C.5

FORWARD 2

Describe unions, intersections, and complements of events as subsets of a sample space • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A card is drawn at random from cards numbered 1 to 10. Event A is "an even number." Event B is "a number greater than 6."

List the outcomes in "A and B" (the intersection).

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q1/5

A card is drawn at random from cards numbered 1 to 10. Event A is "an even number." Event B is "a number greater than 6."

List the outcomes in "A or B" (the union).

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q2/5

A card is drawn at random from cards numbered 1 to 10. Event A is "an even number."

Which set of outcomes is the complement of A ("not A")?

A. 1, 3, 5, 7, 9

B. 2, 4, 6, 8, 10

C. 7, 8, 9, 10

D. 1, 2, 3, 4, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q3/5

True or false?

A card is drawn at random from cards numbered 1 to 10. Event A is "an even number" and event B is "a number greater than 6." Then P(A or B) = $\frac{7}{10}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q4/5

True or false?

A card is drawn at random from cards numbered 1 to 10. Event A is "an even number" and event B is "a number greater than 6." The intersection of A and B has more outcomes than their union.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q5/5

SET 55 • 7.SP.C.6 • STATISTICS & PROBABILITY

Estimate a probability from relative frequency and make predictions

MAIN	Grade 7 main question	**7.SP.C.6**
BACKWARD 1	Use a rate to make a prediction	**6.RP.A.3.b**
BACKWARD 2	Compare decimals to thousandths	**5.NBT.A.3.b**
FORWARD 1	Use relative frequencies in a two-way table to make predictions…	**8.SP.A.4**
FORWARD 2	Find conditional probabilities from a two-way table	**HSS-CP.A.4**

SET 55 • 7.SP.C.6

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.6

A spinner landed on green 36 times in 120 spins.

Estimate the probability of landing on green.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q1/5

A basketball player made 42 of her last 60 free throws.

Predict how many she will make in her next 150 free throws.

Free throws: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q2/5

A coin was flipped 200 times and landed heads 94 times.

What is the relative frequency of heads?

A. 0.94

B. 0.47

C. 0.53

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q3/5

True or false?

A fair number cube (numbered 1 to 6) is rolled 300 times. About 50 of the rolls should be a 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q4/5

True or false?

A spinner landed on red 9 out of 10 times. The probability of red must be exactly 0.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q5/5

SET 55 • 7.SP.C.6

BACKWARD 1

Use a rate to make a prediction • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

A team scores 3 goals every 2 games.

At this rate, how many goals does it score in 10 games?

Goals: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q1/5

4 out of every 10 apples in a crate are green.

At this rate, how many of 250 apples are green?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q2/5

A machine fills 6 bottles every 4 seconds.

At this rate, how many bottles does it fill in 30 seconds?

A. 45

B. 20

C. 24

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q3/5

True or false?

At a rate of 2 out of every 5, there are 40 out of every 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q4/5

True or false?

At a rate of 3 out of every 8, there are 30 out of every 64.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 7.SP.C.6

BACKWARD 2

Compare decimals to thousandths • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b

Which is greater, 0.47 or 0.5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q1/5

Write >, <, or =.

0.305 ___ 0.35

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q2/5

Which decimal is the greatest?

A. 0.059

B. 0.62

C. 0.506

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q3/5

True or false?

$0.250 = 0.25$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q4/5

True or false?

$0.09 > 0.1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 7.SP.C.6

FORWARD 1

Use relative frequencies in a two-way table to make predictions • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows the results of a survey.

Among teens, what is the relative frequency of liking rain?

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q1/5 • NEAREST RELATED

Use the adults row of the table.

Predict how many of 200 adults like rain.

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

Adults: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows the results of a survey.

Which group has the greater relative frequency of liking rain?

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

A. Adults

B. Teens

C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of liking rain among all 100 people is 0.27.

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of disliking rain among adults is 0.42.

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 55 • 7.SP.C.6

FORWARD 2

Find conditional probabilities from a two-way table • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

The table shows students' favorite pets. A student is chosen at random.

Find P(dog | Grade 9).

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q1**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q1/5

The table shows students' favorite pets. A student is chosen at random.

Find P(Grade 10 | cat).

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q2/5

The table shows students' favorite pets. A student is chosen at random.

Find P(cat).

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

A. $\frac{1}{2}$ **B.** $\frac{9}{16}$ **C.** $\frac{7}{16}$ **D.** $\frac{3}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q3/5

True or false?

In the table, P(dog | Grade 10) = $\frac{1}{2}$.

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q4/5

True or false?

In the table, P(Grade 9 | dog) = $\frac{25}{40}$.

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q5/5

SET 70 • 7.SP.C.6 • STATISTICS & PROBABILITY

Collect data from a chance process and use it to estimate a probability

MAIN	Grade 7 main question	**7.SP.C.6**
BACKWARD 1	Organize data with up to four categories	**2.MD.D.10**
BACKWARD 2	Write fractions with denominators of 10 or 100 as decimals	**4.NF.C.6**
FORWARD 1	Record results of two chance processes in a two-way table — near…	**8.SP.A.4**
FORWARD 2	Use collected data to judge whether two events seem independe…	**HSS-CP.A.2**

SET 70 • 7.SP.C.6

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.6

Use these random digits as rolls of a fair number cube. Use only the digits 1 to 6, and skip 0, 7, 8, and 9.

4 8 2 6 6 1 9 3 5 6 0 2 4 6 1 7 3 6 5 2

Record the first 15 rolls. What is the relative frequency of rolling a 6?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 70 • MAIN • Q1/5

A student records 20 flips of a coin:

H T T H H H T H T T H H T H H T H H T H

What is the relative frequency of heads? Use it to predict the number of heads in 200 flips.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 70 • MAIN • Q2/5

Roll a number cube (or use a number-cube app) 30 times. Record each result in a tally table. Then find the relative frequency of rolling an even number.

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 70 • MAIN • Q3/5

A student records 50 spins of a spinner: 18 red, 22 blue, and 10 green.

What is the relative frequency of blue?

A. 0.36

B. 0.44

C. 0.22

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 70 • MAIN • Q4/5

True or false?

A student rolls a fair number cube 30 times. Her data must show exactly 5 threes, or she made a mistake.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 70 • MAIN • Q5/5

SET 70 • 7.SP.C.6

BACKWARD 1

Organize data with up to four categories • Grade 2 • Measurement & Data • 2.MD.D.10

A spinner gives these results: R, B, R, G, R, B, R, G, B, R.

How many times did it land on R?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

Grade 2 • Measurement & Data • 2.MD.D.10 • Set 70 • BACKWARD 1 • Q1/5

A spinner gives these results: R, B, R, G, R, B, R, G, B, R.

How many results are there in all?

Results: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

Grade 2 • Measurement & Data • 2.MD.D.10 • Set 70 • BACKWARD 1 • Q2/5

A tally chart shows red 4, blue 3, and green 3.

How many results are there in all?

A. 7

B. 4

C. 10

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

Grade 2 • Measurement & Data • 2.MD.D.10 • Set 70 • BACKWARD 1 • Q3/5

True or false?

A tally chart can be used to record results as they happen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

Grade 2 • Measurement & Data • 2.MD.D.10 • Set 70 • BACKWARD 1 • Q4/5

True or false?

Red has 5 tally marks and blue has 3. Blue has 2 more than red.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

Grade 2 • Measurement & Data • 2.MD.D.10 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • 7.SP.C.6

BACKWARD 2

Write fractions with denominators of 10 or 100 as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{6}{10}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 70 • BACKWARD 2 • Q1/5

Write $\frac{44}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 70 • BACKWARD 2 • Q2/5

Which decimal is equal to $\frac{35}{100}$?

A. 3.5

B. 0.035

C. 35

D. 0.35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 70 • BACKWARD 2 • Q3/5

True or false?

$\frac{8}{10} = 0.8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 70 • BACKWARD 2 • Q4/5

True or false?

$\frac{5}{100} = 0.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • 7.SP.C.6

FORWARD 1

Record results of two chance processes in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A coin was flipped and a number cube rolled 40 times. The table shows the results.

What is the relative frequency of heads?

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 70 • FORWARD 1 • Q1/5 • NEAREST RELATED

A coin was flipped and a number cube rolled 40 times. The table shows the results.

Among the heads results, what is the relative frequency of an even number?

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 70 • FORWARD 1 • Q2/5 • NEAREST RELATED

A coin was flipped and a number cube rolled 40 times. The table shows the results.

Which result happened most often?

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

A. Tails and odd

B. Tails and even

C. Heads and even

D. Heads and odd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 70 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of tails and odd is 0.3.

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 70 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, the relative frequencies of the four results add to more than 1.

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 70 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 70 • 7.SP.C.6

FORWARD 2

Use collected data to judge whether two events seem independent • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

A coin was flipped and a number cube rolled 40 times. The table shows the results.

Estimate P(heads), P(even), and P(heads and even).

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

Estimates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q1**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 70 • FORWARD 2 • Q1/5

A coin was flipped and a number cube rolled 40 times. The table shows the results.

Is P(heads and even) close to P(heads) × P(even)? What does this suggest?

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q2**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 70 • FORWARD 2 • Q2/5

Events A and B are independent.

Which statement must be true?

A. $P(A) = P(B)$

B. $P(A \text{ and } B) = P(A) + P(B)$

C. $P(A \text{ or } B) = 1$

D. $P(A \text{ and } B) = P(A) \times P(B)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 70 • FORWARD 2 • Q3/5

True or false?

A fair coin is flipped and a fair number cube numbered 1 to 6 is rolled. P(heads and 6) = $\frac{1}{12}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 70 • FORWARD 2 • Q4/5

True or false?

Events A and B are independent, with P(A) = 0.5 and P(B) = 0.4. Then P(A and B) = 0.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 70 • FORWARD 2 • Q5/5

SET 56 • 7.SP.C.7.a • STATISTICS & PROBABILITY

Develop a uniform probability model and use it to find probabilities

MAIN	Grade 7 main question	**7.SP.C.7.a**
BACKWARD 1	Write equivalent fractions	**4.NF.A.1**
BACKWARD 2	Write a fraction for part of a whole	**3.NF.A.1**
FORWARD 1	Use a function that assigns a probability to each outcome — near…	**8.F.A.1**
FORWARD 2	Use the addition rule for probability	**HSS-CP.B.7**

SET 56 • 7.SP.C.7.a

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.7.a

A fair number cube has faces numbered 1 to 6.

What is the probability of rolling a number greater than 4?

Probability: _____

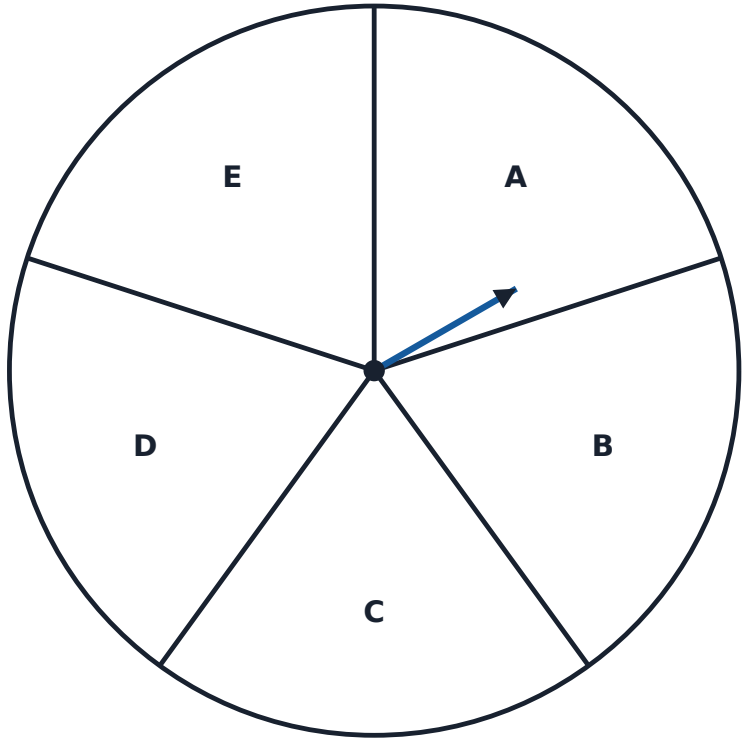
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q1/5

The spinner has 5 equal sections.

What is the probability of landing on a vowel?



Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q2/5

A class has 12 girls and 18 boys. One student is chosen at random.

What is the probability that a girl is chosen?

A. $\frac{2}{5}$

B. $\frac{2}{3}$

C. $\frac{1}{12}$

D. $\frac{3}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q3/5

True or false?

In a uniform probability model, every outcome has the same probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q4/5

True or false?

A letter is chosen at random from the word MATH.

The probability of choosing a vowel is $\frac{1}{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q5/5

SET 56 • 7.SP.C.7.a

BACKWARD 1

Write equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write a fraction equivalent to $\frac{3}{12}$ with a denominator of 4.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q1/5

Write a fraction equivalent to $\frac{8}{12}$ with a denominator of 3.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q2/5

Which fraction is equivalent to $\frac{4}{6}$?

A. $\frac{2}{3}$

B. $\frac{6}{4}$

C. $\frac{4}{12}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q3/5

True or false?

$\frac{5}{10} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q4/5

True or false?

$\frac{2}{8} = \frac{1}{3}$

A. True

B. False

SET 56 • 7.SP.C.7.a

BACKWARD 2

Write a fraction for part of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

A spinner has 8 equal parts. 3 of the parts are red.

What fraction of the spinner is red?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q1/5

A bag has 6 equal-size tiles. 1 tile is blue.

What fraction of the tiles is blue?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q2/5

A pizza is cut into 4 equal slices. 3 slices are eaten.

What fraction of the pizza is eaten?

A. $\frac{3}{6}$

B. $\frac{4}{3}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q3/5

True or false?

One of 6 equal parts is $\frac{1}{6}$ of the whole.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q4/5

True or false?

Two of 6 equal parts is $\frac{2}{4}$ of the whole.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • 7.SP.C.7.a

FORWARD 1

Use a function that assigns a probability to each outcome • Grade 8 • Functions • 8.F.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A spinner has 4 equal sections, A, B, C, and D, and a fair pointer. A probability model assigns a probability to each outcome.

Write the model as pairs (outcome, probability). Is it a function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q1/5 • NEAREST RELATED

A rule assigns (red, 0.5), (blue, 0.3), and (red, 0.2).

Is this rule a function from outcomes to probabilities? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q2/5 • NEAREST RELATED

A fair number cube numbered 1 to 6 is rolled. The function P gives the probability of each outcome.

What is P(4)?

A. $\frac{4}{6}$

B. $\frac{1}{6}$

C. $\frac{1}{4}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A probability model gives exactly one probability to each outcome, just as a function gives exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In a function, one input can have two different outputs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 56 • 7.SP.C.7.a

FORWARD 2

Use the addition rule for probability • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

A card is chosen at random from cards numbered 1 to 10.

Find P(even or greater than 7).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q1/5

P(A) = 0.4, P(B) = 0.5, and P(A and B) = 0.2.

Find P(A or B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q2/5

A fair number cube numbered 1 to 6 is rolled.

What is P(less than 3 or odd)?

A. $\frac{5}{6}$

B. $\frac{2}{3}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q3/5

2025

True or false?

If A and B cannot both happen, then P(A or B) = P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q4/5

True or false?

P(A or B) is always equal to P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q5/5

SET 57 • 7.SP.C.7.b • STATISTICS & PROBABILITY

Develop a probability model from observed frequencies

MAIN	Grade 7 main question	**7.SP.C.7.b**
BACKWARD 1	Read a scaled bar graph	**3.MD.B.3**
BACKWARD 2	Write a part of a whole as a percent	**6.RP.A.3.c**
FORWARD 1	Construct a two-way table from survey counts — nearest related…	**8.SP.A.4**
FORWARD 2	Summarize categorical data with relative frequencies	**HSS-ID.B.5**

SET 57 • 7.SP.C.7.b

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.7.b

A bottle cap is tossed 80 times. It lands top-up 52 times.

Estimate the probability that it lands top-up.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q1/5

A spinner with three unequal sections is spun 200 times: red 90 times, blue 70 times, and green 40 times.

Write an estimated probability for each color.

Probabilities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q2/5

A spinner landed on red 90 times, blue 70 times, and green 40 times in 200 spins.

Which is the best prediction for 1,000 spins?

A. About 900 reds

B. About 333 reds

C. About 90 reds

D. About 450 reds

True or false?

A probability estimated from observed frequencies can change when more trials are done.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q4/5

True or false?

A model built from 10 tosses is just as reliable as a model built from 1,000 tosses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q5/5

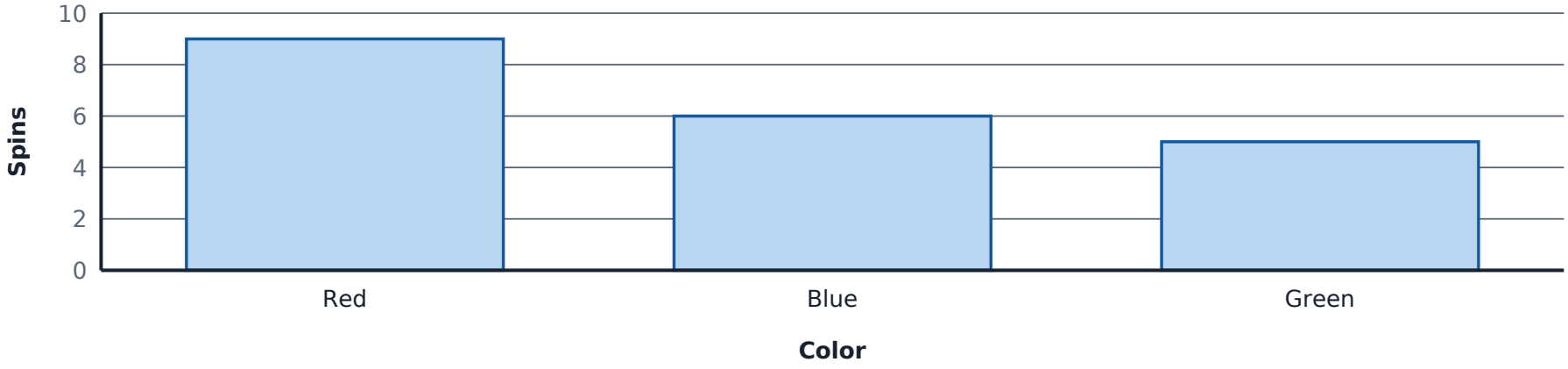
SET 57 • 7.SP.C.7.b

BACKWARD 1

Read a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

The bar graph shows the results of some spins.

How many spins landed on red?



Spins: _____

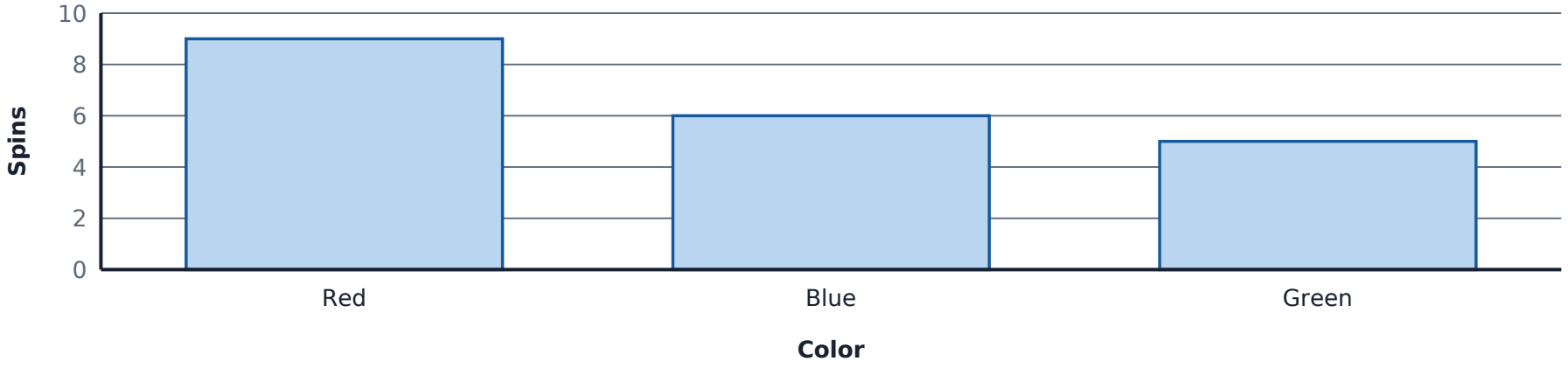
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q1/5

The bar graph shows the results of some spins.

How many more spins landed on red than on green?



Spins: _____

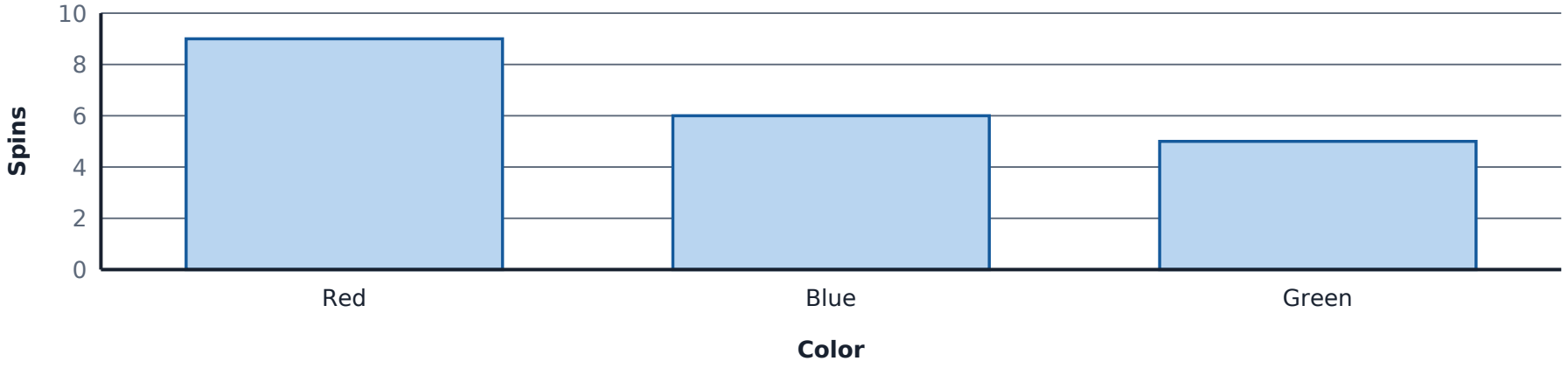
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q2/5

The bar graph shows the results of some spins.

How many spins are shown in all?



A. 9 **B.** 11 **C.** 15 **D.** 20

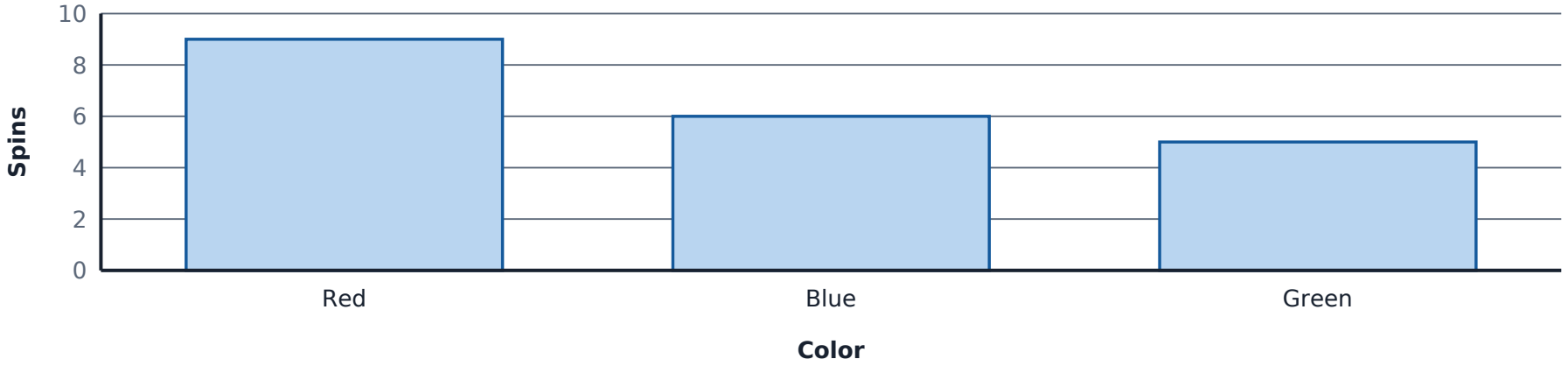
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q3/5

True or false?

The bar graph shows that blue was spun 6 times.



A. True

B. False

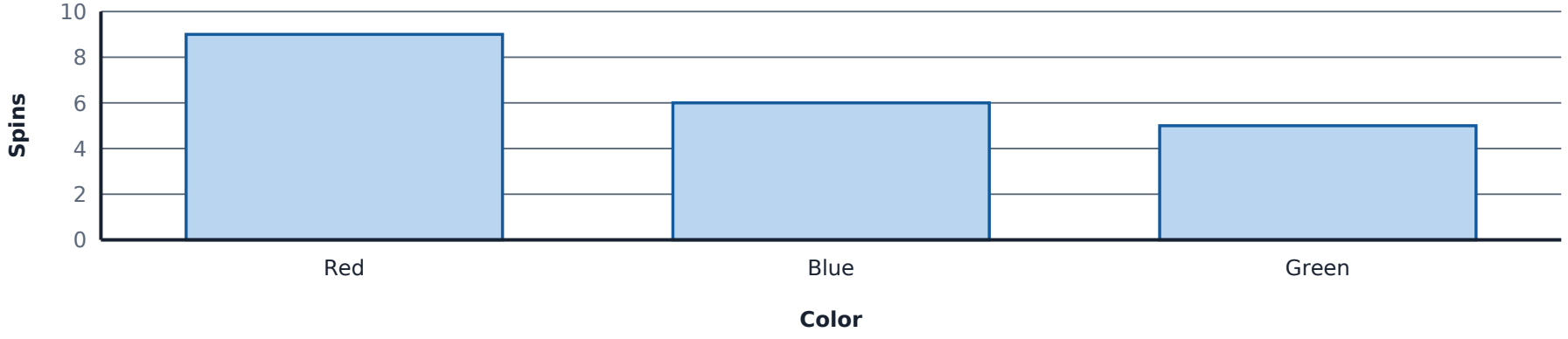
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q4/5

True or false?

The bar graph shows that green was spun more times than blue.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • 7.SP.C.7.b

BACKWARD 2

Write a part of a whole as a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write 45 out of 60 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q1/5

Write 13 out of 50 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q2/5

What percent is 72 out of 120?

A. 72%

B. 48%

C. 60%

D. 40%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q3/5

True or false?

9 out of 36 is 25%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q4/5

True or false?

14 out of 40 is 14%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 7.SP.C.7.b

FORWARD 1

Construct a two-way table from survey counts • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

50 students were surveyed. 30 have a phone. 18 of the students with a phone also have a tablet. 8 of the students without a phone have a tablet.

Complete the two-way table.

	Tablet	No tablet	Total
Phone			
No phone			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q1/5 • NEAREST RELATED

40 people were surveyed. 15 have a dog. 6 of the dog owners also have a cat. 10 of the people without a dog have a cat.

Complete the two-way table.

	Cat	No cat	Total
Dog			
No dog			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q2/5 • NEAREST RELATED

60 students were surveyed. 35 play a sport. 20 of the sport players also play an instrument. 30 of the students play an instrument.

Complete the two-way table.

	Instrument	No instrument	Total
Sport			
No sport			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q3/5 • NEAREST RELATED

80 adults were surveyed. 48 drink coffee. 12 of the coffee drinkers also drink tea. 20 adults drink neither coffee nor tea.

Complete the two-way table.

	Tea	No tea	Total
Coffee			
No coffee			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q4/5 • NEAREST RELATED

100 students were surveyed: 45 in Grade 7 and 55 in Grade 8. 27 of the Grade 7 students and 22 of the Grade 8 students ride the bus.

Complete the two-way table.

	Bus	No bus	Total
Grade 7			
Grade 8			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 57 • 7.SP.C.7.b

FORWARD 2

Summarize categorical data with relative frequencies • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

The table shows the results of a survey.

What is the joint relative frequency of Grade 9 students who play music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q1/5

The table shows the results of a survey.

What is the marginal relative frequency of students who do not play music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q2/5

The table shows the results of a survey.

What is the conditional relative frequency of playing music among Grade 10 students?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. 0.7 **B.** 0.4 **C.** 0.3 **D.** 0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q3/5

True or false?

A marginal relative frequency uses a row total or a column total divided by the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q4/5

True or false?

A conditional relative frequency always divides by the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q5/5

SET 58 • 7.SP.C.7 • STATISTICS & PROBABILITY

Compare probabilities from a model with observed frequencies and explain any differences

MAIN	Grade 7 main question	**7.SP.C.7**
BACKWARD 1	Find a fraction of a whole number	**5.NF.B.4.a**
BACKWARD 2	Divide whole numbers to find equal shares	**4.NBT.B.6**
FORWARD 1	Compare a linear model's predictions with observed data — near…	**8.SP.A.3**
FORWARD 2	Judge whether a result is consistent with a model by using a simu…	**HSS-IC.A.2**

SET 58 • 7.SP.C.7

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.7

A fair number cube is rolled 60 times, and a 5 comes up 14 times.

How many 5s does the model predict? How does the observed count compare?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q1/5

A coin is claimed to be fair. It is flipped 100 times and lands heads 61 times.

Compare this with the model. Give a possible reason for the difference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q2/5

A spinner has 4 equal sections, one of them yellow. In 80 spins, it lands on yellow 35 times.

Which conclusion is best?

A. The model predicts 40 yellows.

B. Yellow came up much more than the expected 20 times, so the spinner may not be fair.

C. Yellow should come up 35 times in every 80 spins.

D. The results match the model exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q3/5

True or false?

A good probability model does not have to match observed frequencies exactly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q4/5

True or false?

A coin that lands heads 7 times in 10 flips must be unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q5/5

SET 58 • 7.SP.C.7

BACKWARD 1

Find a fraction of a whole number • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Find the value.

$\frac{1}{6} \times 60$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

What is $\frac{1}{4}$ of 80?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

What is $\frac{2}{5}$ of 50?

A. 25

B. 125

C. 20

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{3}$ of 90 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q4/5

True or false?

$\frac{3}{4}$ of 40 is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • 7.SP.C.7

BACKWARD 2

Divide whole numbers to find equal shares • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Divide.

240 ÷ 4

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q1/5

Divide.

$300 \div 6$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q2/5

Divide.

1,000 ÷ 8

A. 105

B. 150

C. 125

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q3/5

True or false?

$500 \div 5 = 100$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q4/5

True or false?

$720 \div 3 = 24$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • 7.SP.C.7

FORWARD 1

Compare a linear model's predictions with observed data • Grade 8 • Statistics & Probability • 8.SP.A.3

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A model is $y = 2x + 1$. At $x = 4$, the observed value is $y = 11$.

How far is the observed value from the prediction?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q1/5 • NEAREST RELATED

A model is $y = 5x$. At $x = 3$, the observed value is $y = 13$.

Is the model's prediction too high or too low, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which observed point is closest to the model $y = x + 3$?

A. (2, 7)

B. (2, 9)

C. (2, 3)

D. (2, 5.2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A linear model can be useful even when the data points do not all lie on the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

If one data point is far from the line of fit, the model must be thrown out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q5/5 • NEAREST RELATED

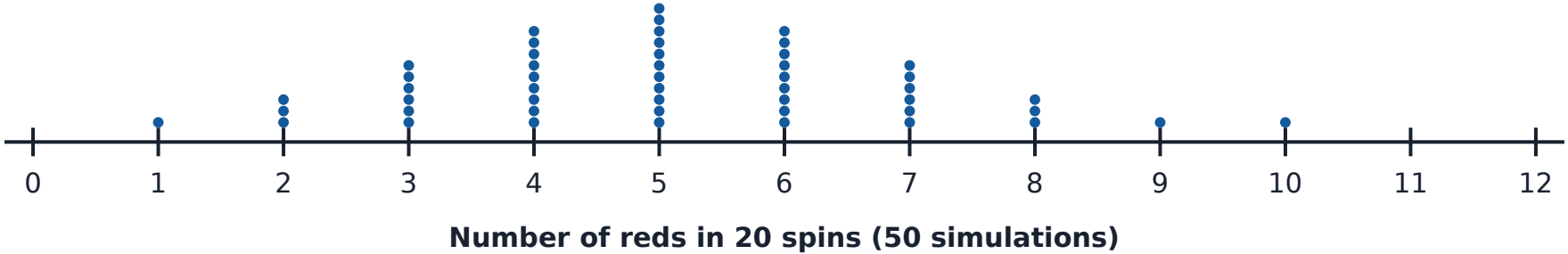
SET 58 • 7.SP.C.7

FORWARD 2

Judge whether a result is consistent with a model by using a simulation • High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A spinner is claimed to land on red 25% of the time. The dot plot shows the number of reds in each of 50 simulated sets of 20 spins under that model. A real spinner lands on red 11 times in 20 spins.

Is the result consistent with the model? Use the simulation.



Answer: _____

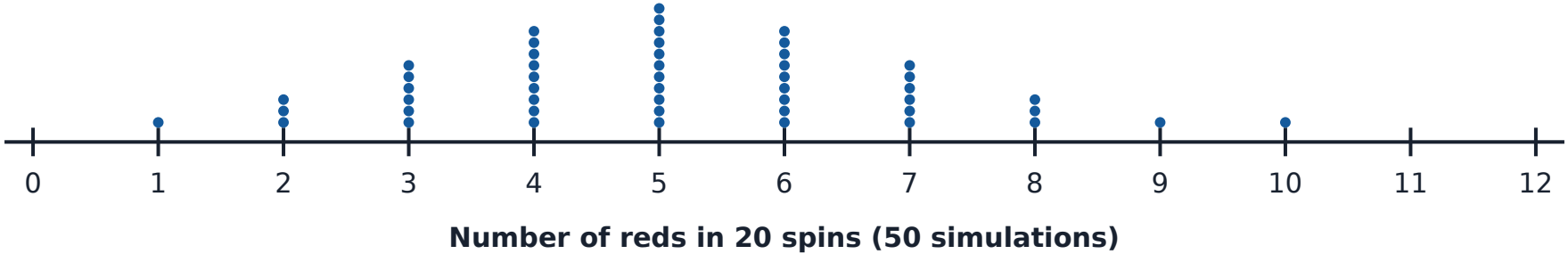
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q1/5

A spinner is claimed to land on red 25% of the time. The dot plot shows the number of reds in each of 50 simulated sets of 20 spins under that model. A real spinner lands on red 6 times in 20 spins.

Is the result consistent with the model? Use the simulation.



Answer: _____

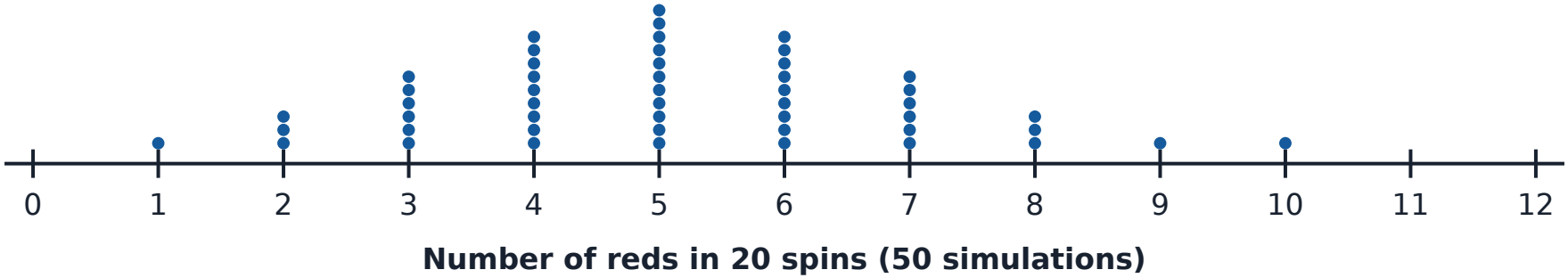
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q2/5

The dot plot shows the number of reds in each of 50 simulated sets of 20 spins of a spinner that lands on red 25% of the time.

Which real result would be the most surprising under this model?



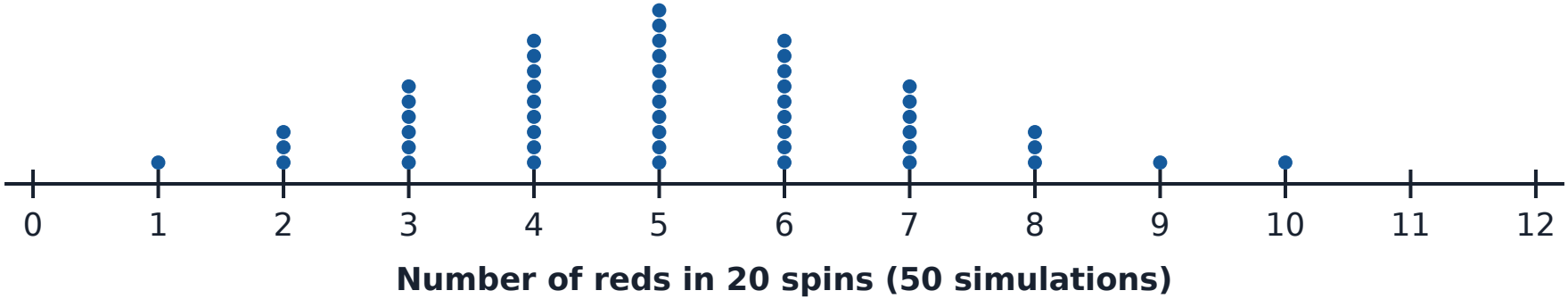
A. 6 reds **B.** 4 reds **C.** 5 reds **D.** 10 reds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q3/5

True or false?

In the simulation shown, 9 or more reds happened in 2 of the 50 sets, about 4% of the time.



A. True

B. False

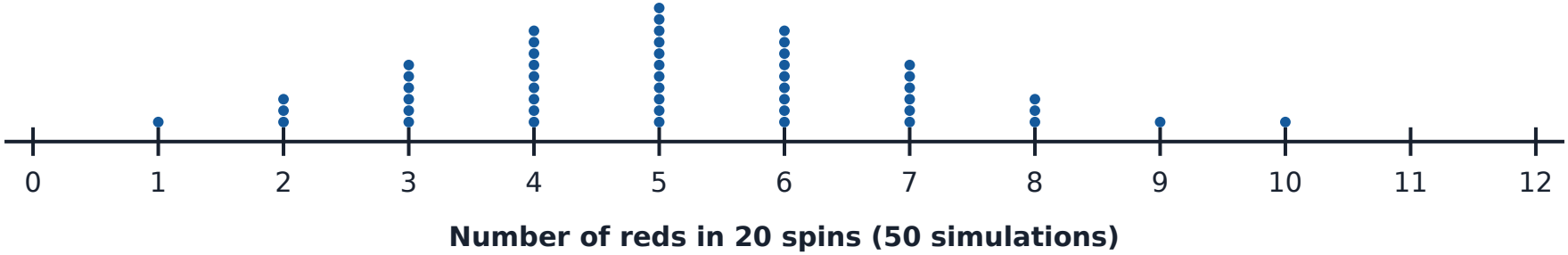
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q4/5

True or false?

A real spinner gives 8 reds in 20 spins. Because 8 is more than the expected 5, the simulation shown proves the 25% model is wrong.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q5/5

SET 59 • 7.SP.C.8.a • STATISTICS & PROBABILITY

Find the probability of a compound event as the fraction of outcomes in the sample space

MAIN	Grade 7 main question	**7.SP.C.8.a**
BACKWARD 1	Multiply fractions	**5.NF.B.4.a**
BACKWARD 2	Count pairs with an array: one row for each first choice	**3.OA.A.3**
FORWARD 1	Find joint relative frequencies in a two-way table — nearest relate…	**8.SP.A.4**
FORWARD 2	Use the multiplication rule for independent events	**HSS-CP.A.2**

SET 59 • 7.SP.C.8.a

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.8.a

A fair coin is flipped and a fair number cube numbered 1 to 6 is rolled.

What is the probability of getting heads and a 6?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q1/5

Two fair coins are flipped.

What is the probability that both land tails?

Probability: _____

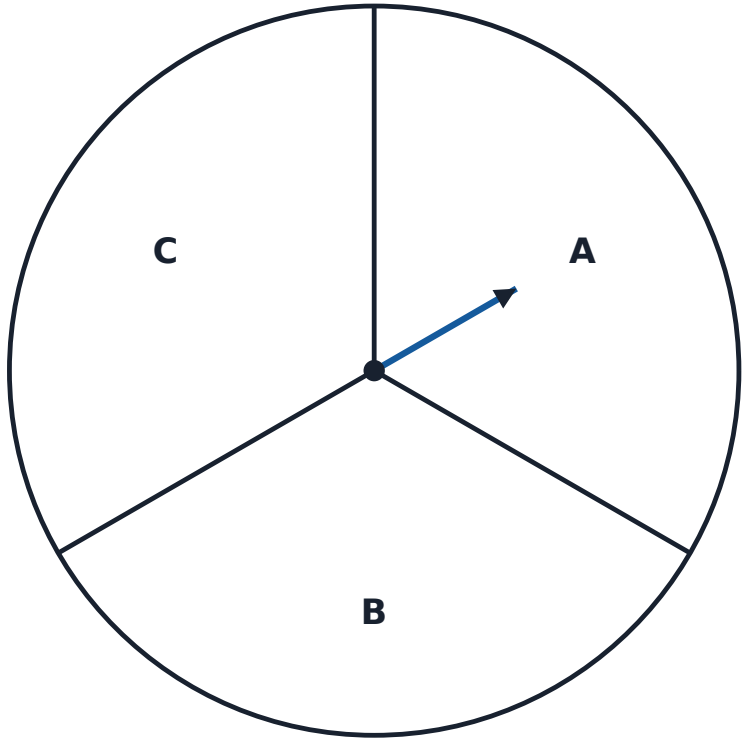
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q2/5

The spinner has 3 equal sections and a fair pointer. It is spun twice, and the spins do not affect each other.

What is the probability of landing on A both times?



A. $\frac{2}{3}$ **B.** $\frac{1}{6}$ **C.** $\frac{1}{9}$ **D.** $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q3/5

True or false?

Two fair number cubes numbered 1 to 6 are rolled. The probability that the sum is 7 is $\frac{6}{36} = \frac{1}{6}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q4/5

True or false?

Two fair coins are flipped. The probability of getting one head and one tail is $\frac{1}{4}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q5/5

SET 59 • 7.SP.C.8.a

BACKWARD 1

Multiply fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$\frac{1}{2} \times \frac{1}{6}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q1/5

Multiply.

$\frac{2}{3} \times \frac{3}{4}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q2/5

Multiply.

$\frac{1}{4} \times \frac{1}{4}$

A. $\frac{2}{4}$

B. $\frac{1}{8}$

C. $\frac{1}{16}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{5} \times \frac{1}{5} = \frac{2}{5}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

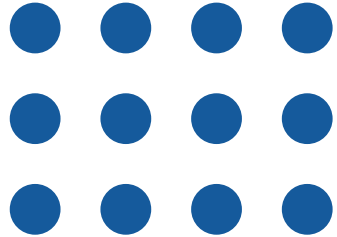
Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q5/5

SET 59 • 7.SP.C.8.a

BACKWARD 2

Count pairs with an array: one row for each first choice • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

A café has 3 kinds of sandwiches and 4 kinds of drinks. The array has one row for each sandwich and one column for each drink; each dot is one sandwich-and-drink pair.



How many pairs are there?

Pairs: _____

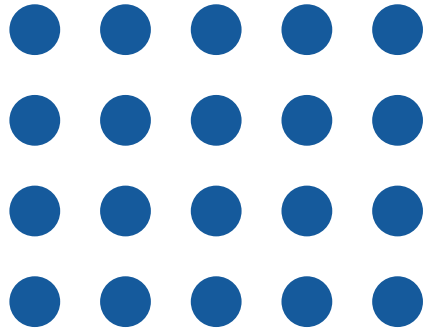
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q1/5

Ty has 4 shirts and 5 pairs of pants. The array has one row for each shirt and one column for each pair of pants; each dot is one outfit.

How many outfits are there?



Outfits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q2/5

A shop has 6 ice cream flavors and 2 kinds of cones. Each flavor can go with each cone, so there are 6 rows with 2 choices in each row.

How many flavor-and-cone choices are there?

A. 12

B. 8

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q3/5

True or false?

3 rows of 7 chairs is 21 chairs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q4/5

True or false?

An array has one row for each of 5 hats and one column for each of 3 scarves. It shows 8 hat-and-scarf pairs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • 7.SP.C.8.a

FORWARD 1

Find joint relative frequencies in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows 100 trials of flipping a coin and rolling a number cube.

What is the relative frequency of an even number and heads?

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows 100 trials of flipping a coin and rolling a number cube.

What is the relative frequency of tails?

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows 100 trials of flipping a coin and rolling a number cube.

Which result happened most often?

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

A. Even and tails

B. Odd and tails

C. Even and heads

D. Odd and heads

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of an odd number and tails is 0.23.

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

The relative frequencies of the four results in the table add to 2.

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 59 • 7.SP.C.8.a

FORWARD 2

Use the multiplication rule for independent events • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

Events A and B are independent. P(A) = 0.3 and P(B) = 0.5.

Find P(A and B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q1/5

A fair coin is flipped 3 times, and the flips are independent.

What is the probability that all 3 flips are heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q2/5

A fair number cube numbered 1 to 6 is rolled twice, and the rolls are independent.

What is the probability of rolling two 6s?

A. $\frac{1}{6}$

B. $\frac{1}{36}$

C. $\frac{1}{12}$

D. $\frac{2}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q3/5

True or false?

Events A and B are independent when P(A and B) = P(A) × P(B).

A. True

B. False

True or false?

Drawing two cards from a deck without putting the first card back are independent events.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q5/5

SET 60 • 7.SP.C.8.b • STATISTICS & PROBABILITY

Represent the sample space of a compound event with a list, a table, or a tree diagram

MAIN	Grade 7 main question	**7.SP.C.8.b**
BACKWARD 1	Use repeated addition to find the total in an array	**2.OA.C.4**
BACKWARD 2	Form ordered pairs from two numerical patterns	**5.OA.B.3**
FORWARD 1	Organize two categorical variables in a two-way table — nearest…	**8.SP.A.4**
FORWARD 2	Find a conditional probability as P(A and B) ÷ P(B)	**HSS-CP.A.3**

SET 60 • 7.SP.C.8.b

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.8.b

A coin is flipped and a spinner with sections 1, 2, and 3 is spun.

List the sample space.

Sample space: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q1/5

A café offers 3 sandwiches (ham, egg, tuna) and 2 drinks (milk, juice).

Make an organized list of every sandwich-and-drink pair.

Sample space: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q2/5

Draw a tree diagram for choosing a shirt (red or blue) and then pants (black, tan, or gray). Write the outcome at the end of each branch.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q3/5

Complete the table to show the sample space for flipping a coin and then rolling a number cube numbered 1 to 6.

	1	2	3	4	5	6
H						
T						

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q4/5

A first coin and a second coin are flipped.

Make an organized list of the sample space. Then list the outcomes in the event "exactly one head."

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q5/5

SET 60 • 7.SP.C.8.b

BACKWARD 1

Use repeated addition to find the total in an array • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

An array has 3 rows with 4 objects in each row.

Write an addition equation for the total.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q1/5

An array has 2 rows with 5 objects in each row.

Write an addition equation for the total.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q2/5

Which equation gives the total of an array with 4 rows of 3?

A. 4 + 3 = 7

B. 3 + 3 + 3 + 3 = 12

C. 3 + 3 + 3 = 9

D. 3 + 4 + 3 = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q3/5

True or false?

An array with 3 rows of 5 has 5 + 5 + 5 = 15 objects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q4/5

True or false?

An array with 2 rows of 4 has 4 + 2 = 6 objects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • 7.SP.C.8.b

BACKWARD 2

Form ordered pairs from two numerical patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 4.

Write the first 3 ordered pairs (A, B).

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q1/5

Pattern A starts at 1 and adds 1. Pattern B starts at 3 and adds 3.

Write the first 3 ordered pairs (A, B).

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q2/5

Pattern A is 0, 3, 6, 9. Pattern B is 0, 6, 12, 18.

Which ordered pair (A, B) comes from these patterns?

A. (3, 3)

B. (12, 6)

C. (6, 9)

D. (6, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q3/5

True or false?

In the patterns A: 0, 5, 10 and B: 0, 10, 20, each term in B is twice the matching term in A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q4/5

True or false?

The patterns A: 0, 2, 4 and B: 0, 3, 6 make the ordered pair (4, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • 7.SP.C.8.b

FORWARD 1

Organize two categorical variables in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

In a class, 14 students have a sibling and 10 do not. 8 of the students with a sibling have a pet. 3 of the students without a sibling have a pet.

Complete the two-way table.

	Pet	No pet	Total
Sibling			
No sibling			
Total			

A class has 30 students. 18 like soccer. 12 of the soccer fans also like basketball. 4 of the students who do not like soccer like basketball.

Complete the two-way table.

	Basketball	No basketball	Total
Soccer			
No soccer			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q2/5 • NEAREST RELATED

50 shoppers were counted. 20 used a coupon. 15 of the coupon users paid by card. 25 of the shoppers paid by card.

Complete the two-way table.

	Card	Cash	Total
Coupon			
No coupon			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q3/5 • NEAREST RELATED

A team played 36 games. It played 20 at home and won 14 of them. It won 22 games in all.

Complete the two-way table.

	Won	Lost	Total
Home			
Away			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q4/5 • NEAREST RELATED

90 students were surveyed. 40 walk to school, and 25 of them eat breakfast at home. 30 of the students who do not walk eat breakfast at home.

Complete the two-way table.

	Breakfast	No breakfast	Total
Walk			
Do not walk			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 60 • 7.SP.C.8.b

FORWARD 2

Find a conditional probability as P(A and B) ÷ P(B) • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 0.12 and P(B) = 0.4.

Find P(A | B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q1/5

Two fair coins are flipped. At least one landed heads.

What is the probability that both landed heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q2/5

A fair number cube numbered 1 to 6 is rolled, and the result is even.

What is the probability that it is a 6?

A. $\frac{1}{6}$

B. $\frac{1}{2}$

C. $\frac{1}{3}$

D. $\frac{2}{3}$

True or false?

If A and B are independent and P(B) > 0, then P(A | B) = P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q4/5

True or false?

P(A | B) is always equal to P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q5/5

SET 61 • 7.SP.C.8.c • STATISTICS & PROBABILITY

Design and use a simulation to estimate the probability of a compound event

MAIN	Grade 7 main question	**7.SP.C.8.c**
BACKWARD 1	Display data in a dot plot	**6.SP.B.4**
BACKWARD 2	Compare fractions with the same numerator or the same denomin…	**3.NF.A.3.d**
FORWARD 1	Construct and interpret a scatter plot — nearest related (no direct…	**8.SP.A.1**
FORWARD 2	Use a simulation to decide whether a difference between treatme…	**HSS-IC.B.5**

SET 61 • 7.SP.C.8.c

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.8.c

A player makes 60% of her free throws, and each shot is independent of the others.

Describe how to use random digits 0 to 9 to simulate one trial of 2 free throws.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q1/5

A player makes 60% of her free throws; each shot is independent. In the simulation, the digits 0 to 5 mean a make and 6 to 9 mean a miss. Each pair of digits is one trial of 2 shots:

32 70 15 86 41 09 55 63 27 84

In how many trials are both shots made? Estimate the probability that both shots are made.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q2/5

Each of 3 children is equally likely to be a boy or a girl, and the children are independent.

Which tool can simulate the 3 children?

A. Spin a spinner with 3 unequal parts once

B. Flip a fair coin 3 times

C. Draw one card from a deck

D. Roll one number cube once

Each of 3 children is independently equally likely to be a boy or a girl. In the simulation, an even digit means a girl and an odd digit means a boy. Each group of 3 digits is one family:

204 317 856 993 120 448 715 362 091 574

Estimate the probability that a family has at least 2 girls.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q4/5

True or false?

A simulation of 3 children uses even digits for girls and odd digits for boys. Its trials are 246, 135, and 802. Exactly 1 of these 3 trials shows at least 2 girls.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q5/5

SET 61 • 7.SP.C.8.c

BACKWARD 1

Display data in a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

Make a dot plot of the numbers of heads in 8 trials:

1, 2, 2, 3, 1, 2, 0, 2



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q1/5

The dot plot shows the number of heads in 10 trials.

In how many trials were there 2 or more heads?



Trials: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q2/5

Which display shows each data value as a dot above a number line?

A. Dot plot

B. Histogram

C. Bar graph

D. Box plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q3/5

True or false?

A dot plot shows every value in a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q4/5

True or false?

A box plot shows every value in a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • 7.SP.C.8.c

BACKWARD 2

Compare fractions with the same numerator or the same denominator • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

Which is greater, $\frac{3}{8}$ or $\frac{5}{8}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q1/5

Write >, <, or =.

$\frac{2}{3}$ ___ $\frac{2}{6}$

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q2/5

Which fraction is less than $\frac{4}{6}$?

A. $\frac{5}{6}$

B. $\frac{6}{6}$

C. $\frac{3}{6}$

D. $\frac{4}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{4} < \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q4/5

True or false?

$\frac{3}{8} > \frac{4}{8}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q5/5

SET 61 • 7.SP.C.8.c

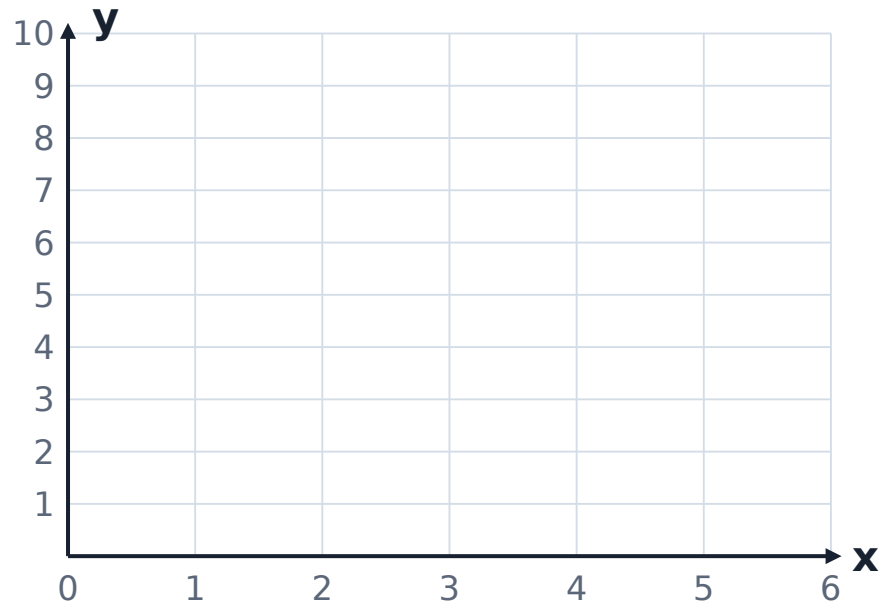
FORWARD 1

Construct and interpret a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Make a scatter plot of the data in the table.

x	1	2	3	4	5
y	2	4	5	7	9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows x-values 1, 2, 3, 4, 5 and y-values 2, 4, 5, 7, 9.

Describe the association between x and y.

Association: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q2/5 • NEAREST RELATED

What does each point in a scatter plot stand for?

A. An outlier

B. One pair of values from the data

C. The mean of the data

D. The line of fit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A scatter plot shows the relationship between two numerical variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

All the points in a scatter plot must lie on a straight line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 61 • 7.SP.C.8.c

FORWARD 2

Use a simulation to decide whether a difference between treatments is significant • High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In an experiment, 40 similar plants were randomly assigned: 20 got a new plant food (Group A) and 20 got none (Group B). Group A grew a mean of 12 cm and Group B a mean of 9 cm. The data were re-randomized 100 times to simulate no treatment effect; a difference of 3 cm or more happened 2 times.

Is there evidence that the plant food made a difference? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q1/5

In the same kind of randomized experiment (plants randomly assigned to plant food or none), the data were re-randomized 100 times to simulate no treatment effect. A difference of 3 cm or more happened 40 times.

What does this suggest about the plant food?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q2/5

Why are the data re-randomized many times in the simulation?

A. To remove outliers

B. To see what differences happen by chance alone

C. To make the mean larger

D. To make the treatment work better

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q3/5

True or false?

Random assignment helps make two treatment groups alike before an experiment begins.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q4/5

True or false?

A difference between two groups in one experiment is always caused by the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q5/5